PAPER · REWRITTEN NOVEMBER 2026

The Illiquidity Premium Changed Hands

Buyers of private funds now pay to wait. The premium goes to whoever supplies the liquidity.

Charlie Brockmeier · 46 min read · 20 charts, 3 tables

First version: May 2025 · Rewritten: November 2026 · Read the original · Companion: The Brockmeier Model

Views are my own and not those of my employer. Nothing here is investment advice. Every figure comes from a public source cited at the end, or is labeled as my own arithmetic; none comes from my employer's data or products, or from any client.

SUMMARY

1. **A ten-year lockup should pay about 2 to 4 points a year. Buyouts have paid 0.1.** Longstaff's bound caps the cost of a ten-year lockup on a buyout-like asset at 5.4% a year, and a realistic investor needs 1.6 to 4.2 points (my arithmetic) [4, 17]. Cambridge Associates' U.S. private equity index beat the same cash flows in the S&P 500 by 2.4 points a year over 25 years and by 0.1 over the last ten [24].

2. **Adjust for leverage and the premium is negative.** Buyout investments carry a market beta of about 1.3 on cash-flow evidence [91] and 1.79 when priced off secondary trades [92]. At 1.3 the 25-year premium becomes –0.1 points a year and the ten-year premium –3.8 (my arithmetic). The index built from secondary prices has an alpha of –2% a year [92].

3. **Investors accept this because smooth marks are worth paying for.** I levered the U.S. stock market 1.3 times and marked it the way an appraiser would, as a four-quarter average. Volatility fell from 22.2% to 11.7%, beta from 1.30 to 0.32 and the 2020 drawdown from 25.9% to 3.2%, with the same assets and the same cash (my arithmetic on public data [93]). AQR now assumes that fees and "investors' preference for smooth returns fully offset any illiquidity premium" in private credit [26].

4. **The same kind of asset clears at five prices.** In 2025-26 a non-traded fund or a continuation vehicle transacted at 100% of NAV [58, 94], an LP buyout stake at 91% [55], public BDCs at about 80% [94], London-listed private equity trusts at 64% to 77% [95, 96], and tender offers for non-traded BDC shares at 70% to 87% [74]. The buyer who pays par is the individual.

5 **The premium still exists. It is paid to the supplier of liquidity.** Buyout stakes changed hands near 54% of NAV in 2009 [59], worth roughly 17% a year to the buyer (my arithmetic). The University of California was guaranteed 11.25% a year for handing a gated fund cash [67]. In 2026, redemption requests at semi-liquid credit funds ran from two to eight times their 5% caps [68–72, 97], and lenders writing new loans that fall got modestly wider spreads and larger equity cushions [100].

My take: the illiquidity premium did not disappear. It changed hands. For the average buyer of a private fund it has been zero or negative for at least a decade once fees and leverage are counted, because enough investors value smooth marks to bid it away, and the semi-liquid money arriving now makes that worse. It is still paid in full to the other side of the trade: whoever holds cash when others are locked, and buys their stakes at a discount or lends when no one else will. So my default allocation to private funds is zero. The exceptions are a discount from a forced seller, a senior loan whose spread over the liquid alternative is contractual and whose fees are low, and a manager whose record beats a levered public benchmark. Section 1 lists what would change my mind.

1. The claim

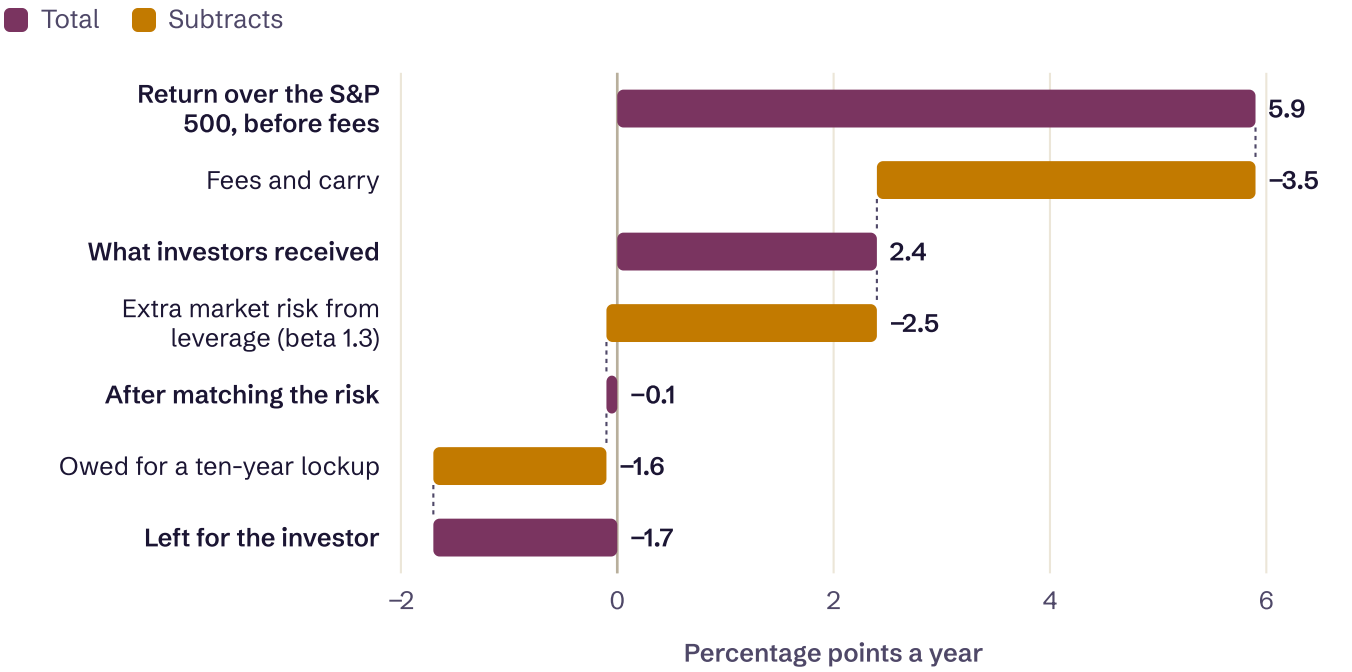
The first two versions of this paper asked how big the illiquidity premium is. This one takes a side. For the average buyer of a private fund the premium is gone, and has been for at least a decade. It did not vanish. It moved to the other side of the trade.

The argument has three steps, and each is a number that can be checked.

1. **What a lockup should pay.** Theory caps the cost of a ten-year lockup on a buyout-like asset at 5.4% a year for a seller with perfect timing, and puts it at 1.6% to 4.2% for realistic investors (Section 2; my arithmetic from Longstaff's bound [4, 17]). In the standard portfolio model with illiquid assets, an investor who would put 59% of a portfolio in an asset that trades freely should hold 5% of it when it trades about once a decade [99].
2. **What it has paid.** Net of fees, the Cambridge Associates U.S. private equity index beat the same cash flows in the S&P 500 by 2.4 points a year over 25 years and by 0.1 over ten [24]. Buyouts are levered, so the fair comparison is a levered index. At the market beta of 1.3 estimated from buyout cash flows [91], the 25-year premium is −0.1 points and the ten-year premium −3.8 (my arithmetic, Section 4). An index built from the prices actually paid for fund stakes has a beta of 1.79 and an alpha of −2% a year [92].
3. **Who collects it now.** A buyer of buyout fund stakes at 54% of NAV in 2009 earned roughly 17 points a year on top of the funds' own return [59]. A university that gave a gated real estate fund $4 billion in 2023 was guaranteed 11.25% a year [67]. Lenders writing new direct loans in the fall of 2026, after a year of redemptions, were getting modestly wider spreads and larger equity cushions than at the end of 2025 [100]. The premium is paid to whoever has cash when others are locked.

Figure 1. Where the premium went: buyouts over the best 25 years on record

Cambridge Associates U.S. private equity index vs. the S&P 500 on the same cash flows, 2001–2025, points a year



My arithmetic. Net premium: 12.4% less 10.0% [24]. Fees: AQR's all-in estimate of 3.5 points a year, which it lowered from 4 this year [26]; my own fee model in Section 7 gives 4.0 to 4.8. Leverage: 0.3 times the S&P 500's return over Treasury bills, 10.0% less 1.72% [91, 93]. Lockup: the patient-investor hurdle from Table 3. Over the last ten years the same steps run 0.1, −3.9, −1.6 and end at −5.4.

Source: Cambridge Associates [24]; AQR [26]; Franzoni, Nowak and Phalippou [91]; Ken French data library [93]; Longstaff [4].

Read Figure 1 from the top. Buyout managers did produce something: about 6 points a year over the S&P 500 before fees, over the best 25-year window on record. Fees took 3.5 of it, on AQR's estimate of the all-in cost [26]. Leverage accounts for the 2.4 that was left. Nothing remains to pay for the lockup, which is worth 1.6 points a year even to a patient investor. Over the last ten years the same ledger ends at −5.4: 0.1 of net premium, less 3.9 for leverage, less 1.6 for the lockup (my arithmetic).

Why anyone accepts that

Because illiquidity is no longer only a cost. An asset appraised four times a year instead of priced every second reports about half the volatility and a quarter of the beta of the same asset in public form. Section 3 shows this on real market data. For a pension board, an endowment committee or an adviser with nervous clients, that is worth paying for. Cliff Asness has argued that investors may now accept a lower return in exchange for not seeing the losses [18]. AQR's 2026 return assumptions say it outright for private credit: "higher fees and investors' preference for smooth returns fully offset any illiquidity premium on average" [26]. I think that is right, and that it holds for buyouts and core real estate too.

Then there is the money. Buyout funds have raised $1.8 trillion since 2022, the industry holds $1.3 trillion of unspent commitments, and about 32,000 unsold companies worth $3.8 trillion are waiting for an exit [63].

Evergreen funds built for individuals held about $607 billion in March 2026 [82]. A premium is a price. This much capital bidding for the same assets is how a price goes to zero.

What would prove me wrong

A hard claim should come with its tests. Any of these would change my mind.

- The Cambridge index beats the S&P 500 by 4 points a year or more over ten years. That is what it takes to cover leverage at a beta of 1.3, about 2.5 points at the 25-year equity premium, plus the 1.6 points a patient investor needs for the lockup.
- Buyout stakes sell below 80% of NAV outside a crisis. At that price the discount alone is worth about 5.7% a year over four years (my arithmetic), and I would be a buyer.
- All-in fees fall from about 3.5 points a year [26] to 2. That alone would move the 25-year ledger from −1.7 to about −0.2.
- Listed private equity trusts and public BDCs trade back at NAV. That would mean the marks are believed by people who are free to sell.
- An index built from transaction prices, not appraisals, shows positive alpha over a full cycle [92].

None of the five is true in November 2026.

2. What a lockup should pay

The 2025 version of this paper defined the illiquidity premium as the extra return an investor should demand for holding an asset that cannot be sold quickly at fair value. That definition holds, but it hides two different things. The first is the cost of illiquidity in normal times: trading costs, the time it takes to sell, the lockup itself. The second is liquidity risk, the chance that an asset becomes hard to sell exactly when its owner most needs cash. Theory says both should be priced. The evidence says the second matters more.

Five models carry most of the weight. Each answers a different question, and private funds sit in the space between them.

Trading costs and clienteles (Amihud and Mendelson, 1986)

If an asset costs S to trade round trip and its holder trades it every 1/μ years, the holder loses about μ·S a year to trading. In equilibrium the asset's expected return has to cover that loss for its marginal holder [1]:

$$E[r_j] - r_f \approx \mu_i \cdot S_j$$

S_j: proportional round-trip trading cost of asset j. μ_i: how often its marginal holder trades, one over the expected holding period.

The useful part is the clientele effect. Long-horizon investors end up holding the expensive-to-trade assets because they spread the cost over many years, so the premium rises with the spread but at a falling rate. On NYSE stocks from 1961 to 1980, a stock with a 1.5% spread earned 0.45 percentage points a month more than one with a

0.5% spread, while a 5% spread stock earned only 0.09 points a month more than a 4% one [2]. That concavity is the theoretical case for endowments. An investor who rarely trades can collect a premium priced for someone who trades more often.

Liquidity as a risk (Acharya and Pedersen, 2005)

The liquidity-adjusted CAPM nets trading costs c out of returns and asks which covariances investors care about [3]:

$$E[r_i - r_f] = E[c_i] + \lambda\,(\beta_{1i} + \beta_{2i} - \beta_{3i} - \beta_{4i})$$

Each β divides a covariance by $\mathrm{var}(r_M - c_M)$. β_1: $\mathrm{cov}(r_i, r_M)$, ordinary market risk. β_2: $\mathrm{cov}(c_i, c_M)$, the asset gets illiquid when the market does. β_3: $\mathrm{cov}(r_i, c_M)$, the asset's return when market liquidity dries up. β_4: $\mathrm{cov}(c_i, r_M)$, the asset's liquidity when the market falls.

The three extra betas all charge for the same bad state: prices fall and nothing trades. An asset that becomes illiquid when the market falls, or falls when market liquidity dries up, needs a higher expected return. Acharya and Pedersen estimate that liquidity risk adds about 1.1 percentage points a year to the return gap between illiquid and liquid U.S. stocks, on top of the level effect [3].

For private markets this is the right lens. A buyout fund's lockup is not the problem in itself. The problem is that the lockup binds in exactly the state the model charges for, when public portfolios are down, distributions stop and capital calls keep coming.

The option to sell (Longstaff, 1995)

Longstaff asked how much the freedom to sell could possibly be worth. Picture an investor with perfect timing who could sell at the best price during a lockup of T years. A lockup costs that investor at most the value of a lookback option on the asset, which has a closed form [4]:

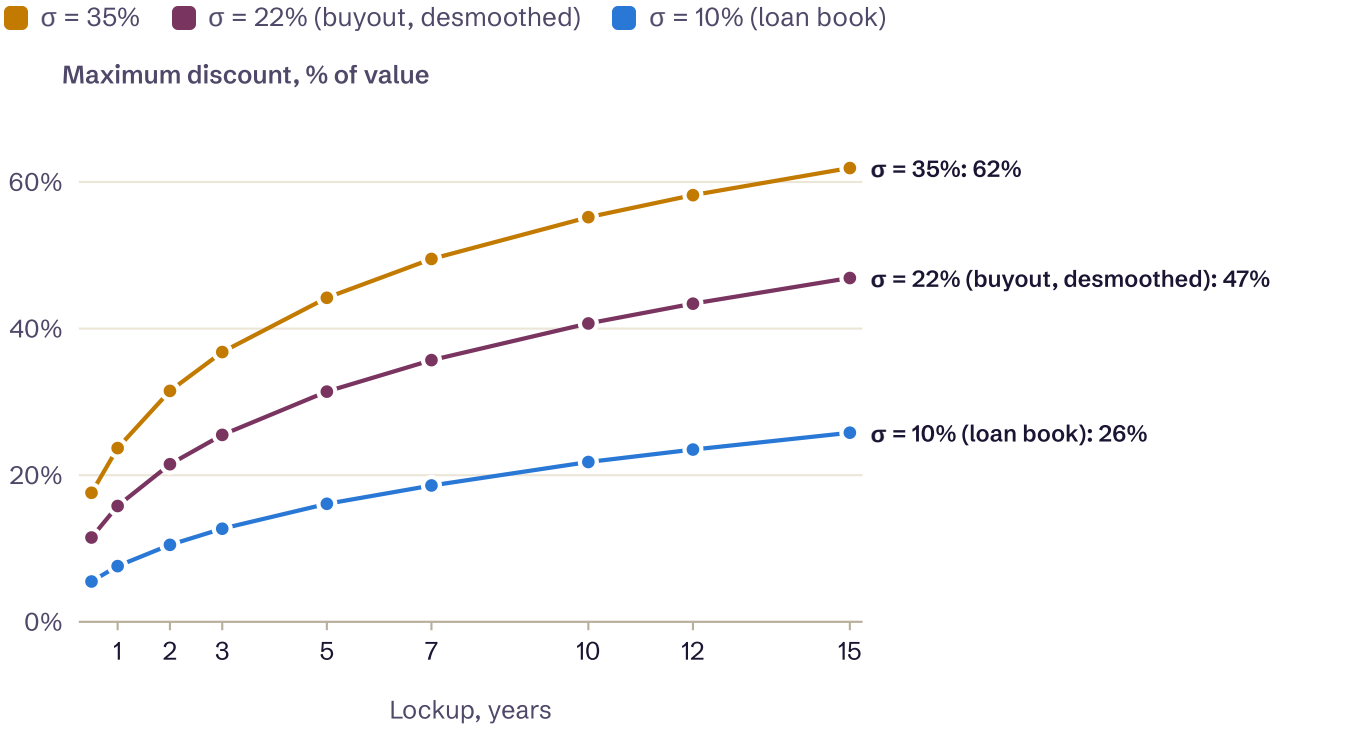
$$F(T, \sigma) = (2 + \sigma^2 T/2) \cdot N\left(\sqrt{(\sigma^2 T)/2}\right) + \sqrt{(\sigma^2 T / 2\pi)} \cdot e^{-\sigma^2 T/8} - 1$$

F: upper bound on the cost of a T-year lockup, as a fraction of the asset's value. σ: annual volatility of the asset. N: the standard normal distribution function.

Two things make this formula useful in diligence. It needs only two inputs, and it is an upper bound, so it tells you where a claimed premium stops being plausible. Figure 2 plots it as a discount to value. A two-year lockup on an asset with 22% volatility, about what a leveraged buyout shows once smoothing is stripped out [17], is worth at most a 21% discount. Ten years is worth at most 41%. Volatility matters as much as time: ten years on a loan book with 10% volatility caps out at 22%.

Figure 2. The most a lockup can cost: Longstaff's upper bound

Maximum discount to value for a perfectly timed seller, by lockup length and asset volatility



My calculation from Longstaff's closed-form bound F, converted to a discount as F / (1 + F). It assumes the seller would have sold at the best price during the lockup, so real costs are lower.

Source: Longstaff (1995) [4]; 22% buyout volatility from Anson (2024) [17].

Spread over the lockup, the bound falls per year as the lockup gets longer, because it grows roughly with the square root of time. For a ten-year fund at 22% volatility it works out to 5.4% a year. That is close to the 3% to 5% that LPs have traditionally asked of private equity over public equity, which makes a useful check: the traditional hurdle is roughly what a perfectly timed seller would need. No one is perfectly timed, so a rational investor needs less. Section 8 turns this into an underwriting rule.

Figure 3. The price of a lockup, per year

Longstaff's upper bound spread over the lockup: the most a perfectly timed seller would pay, % of value a year

Volatility	Lockup, years						
	1	2	3	5	7	10	12
5%	4.1%	2.8%	2.3%	1.8%	1.5%	1.3%	1.1%
10% (loan book)	8.2%	5.7%	4.6%	3.6%	3.0%	2.5%	2.3%
15%	12.5%	8.7%	7.0%	5.3%	4.5%	3.7%	3.4%
22% (buyout)	18.8%	12.8%	10.3%	7.8%	6.5%	5.4%	4.9%
30%	26.3%	17.7%	14.1%	10.6%	8.8%	7.2%	6.5%
40%	36.1%	24.0%	19.0%	14.1%	11.7%	9.5%	8.5%

My calculation: $(1 + F)^{1/T} - 1$, where F is Longstaff's bound. The outlined cell is the standard case in this paper, a ten-year buyout fund at its desmoothed volatility. A real investor uses only part of the option, so Table 3 scales these down.

Source: Longstaff (1995) [4]; 22% buyout volatility from Anson (2024) [17].

Figure 3 gives the same bound per year for every combination. Two things stand out. Short lockups are the expensive ones: at 22% volatility a one-year lockup costs up to 18.8% a year and a ten-year lockup 5.4%, which is why a quarterly gate on a volatile asset is a bigger concession than it looks. And volatility matters as much as time. Ten years on a loan book at 10% volatility costs at most 2.5% a year, which is why private credit is where a premium is easiest to earn.

Liquidity risk as a factor (Pástor and Stambaugh, 2003)

Pástor and Stambaugh measured market-wide liquidity from return reversals. When liquidity is poor, order flow pushes prices further and they bounce back harder the next day. For each stock in each month they estimate [5]:

$$r^{e}_{i,d+1} = \theta_i + \varphi_i r_{i,d} + \gamma_i \cdot sign(r^{e}_{i,d}) \cdot v_{i,d} + \varepsilon_{i,d+1}$$

r^{e}: daily return in excess of the market. v: dollar volume. γ: the reversal per dollar of signed volume, more negative when the stock is less liquid. Averaging γ across stocks gives aggregate liquidity, and a stock's liquidity beta is its exposure to shocks in that aggregate.

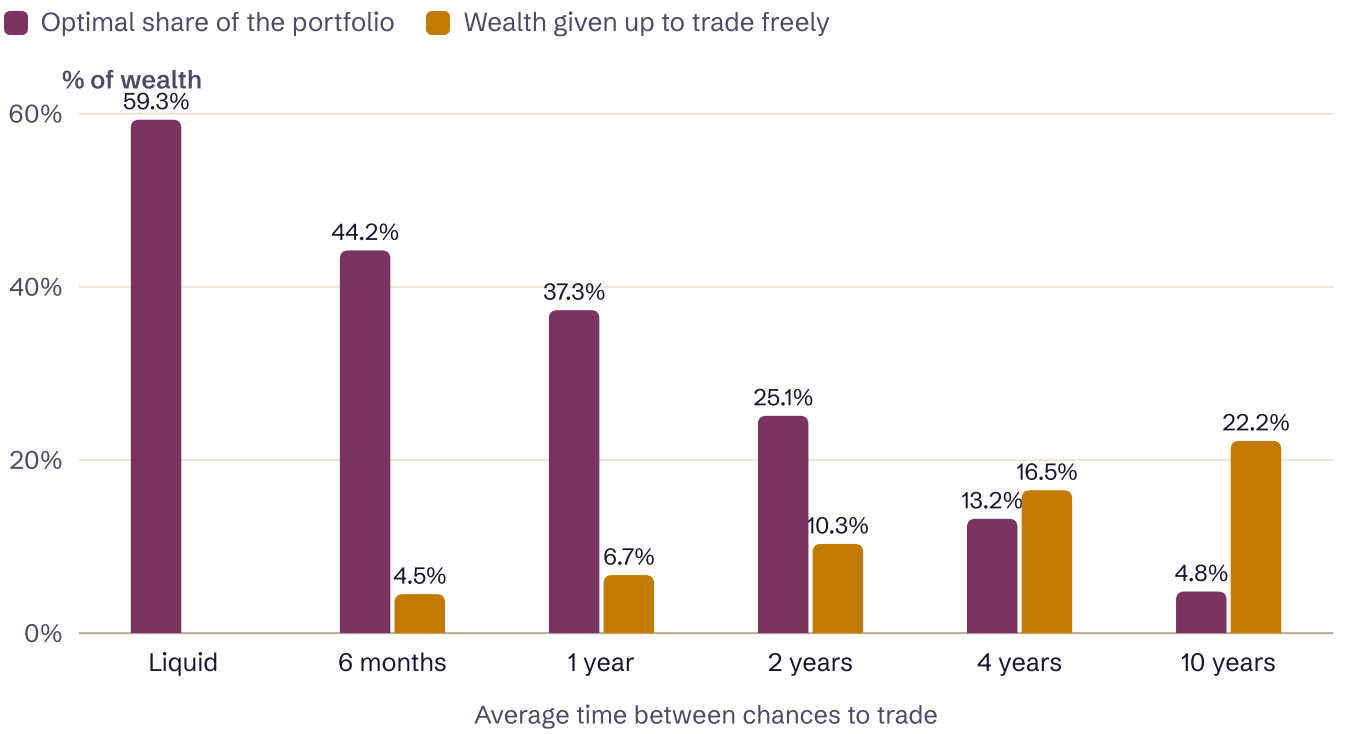
Stocks with high liquidity betas earned 7.5% a year more than stocks with low ones from 1966 to 1999, after adjusting for market, size, value and momentum [5]. Twenty years later two independent replications reproduced the measure, and both found the premium larger after 1999 than before [6]. Few anomalies have held up that well out of sample.

How little to hold (Ang, Papanikolaou and Westerfield, 2014)

The models above price the lockup. This one sizes the position. Ang, Papanikolaou and Westerfield solve the portfolio problem for an investor who can trade one risky asset freely and the other only at random moments, on average once every $1/\lambda$ years [99]. Their baseline gives the two assets the same expected return and risk, so the illiquid one carries no premium at all, and asks how much of it to own.

Figure 4. How much of an illiquid asset to hold when it pays no premium

Optimal share of the portfolio, and the wealth an investor would give up to trade freely, by average time between chances to trade



Baseline model: both risky assets have a 12% expected return and 15% volatility and are uncorrelated, the risk-free rate is 4% and risk aversion is 6. The illiquid asset earns no premium over the liquid one.

Source: Ang, Papanikolaou and Westerfield (2014), Table 3 [99].

If the second asset traded freely the investor would put 59.3% of the portfolio in it. If it trades once a year on average, 37.3%. Once every two years, 25.1%. Once every ten years, 4.8%. The investor would give up 22.2% of total wealth to make a once-a-decade asset tradable [99]. The cost comes from spending, not from rebalancing: consumption has to be paid out of liquid wealth, so an investor who is rich on paper and short of cash is poor in the only way that matters. Uncertainty about when the next exit comes does most of the damage [99].

That result is the benchmark for everything after it. An asset with no premium that trades once a decade deserves about a twelfth of the weight it would get if it were liquid. To hold more, the investor has to be paid.

Put together, the five models say one thing. The level of illiquidity earns a modest premium that shrinks as markets get cheaper to trade. Exposure to liquidity crashes earns a larger and more durable one. Private funds

carry both, plus a feature none of these models covers: the investor cannot see the price. Section 3 shows why that feature has become something investors pay for.

3. Measuring it is the hard part

In public markets the premium is a regression. In private markets it is an argument about measurement, and most of the disagreement in the literature is really about which yardstick to use. Four problems matter: the return metric, stale prices, leverage and factor exposure, and fees. Fees get their own section later.

IRR, TVPI, PME and direct alpha

Funds report two numbers. The internal rate of return (IRR) is the discount rate that sets the fund's cash flows to zero. Total value to paid-in (TVPI) is distributions plus remaining NAV, divided by capital called. Neither says whether the investor beat what a public portfolio would have earned on the same cash flows. Kaplan and Schoar's public market equivalent (PME) does, by growing every cash flow at the realized return of a public index [13]:

$$PME = \Sigma_t D_t \cdot (I_T / I_t) \div \Sigma_t C_t \cdot (I_T / I_t)$$

D_t: distributions, with ending NAV counted as a final distribution. C_t: capital calls. I_t: level of the public index. A PME above 1.0 means the fund beat the index.

Direct alpha takes the same index-compounded cash flows and solves for their IRR, which turns the ratio into an annual excess return [14]. A PME above one and a positive direct alpha say the same thing. I use direct alpha when I want a number in percent a year to compare with a hurdle, and PME when I want to know how many dollars ahead an LP ended up.

Why IRR misleads

IRR rewards timing, not wealth. Take a fund that calls $100 and returns $200 five years later: a 14.9% IRR and a 2.0x multiple. Now let it bridge the first year with a subscription credit line at 6%, so the LP wires the money a year later and the fund pays $6 of interest. The LP now gets $194, four years after paying in. The multiple falls to 1.94x and the IRR rises to 18.0%. The investor is poorer and the headline number is better. (The example is my own arithmetic.)

Across real funds the effect is smaller but runs one way. Schillinger, Braun and Cornel simulated subscription-line use across 6,353 buyout deals in 700 funds and found that drawing a line for six months or less raised the median net IRR by 0.20 points and the mean by 0.47 points, with multiples barely changed [15]. When I diligenced funds I asked for IRR with and without the facility. A manager who would not show both was telling me something.

Stale prices and smoothed returns

Private assets are marked by appraisal, usually quarterly, and appraisers anchor on the last mark. Getmansky, Lo and Makarov model a reported return as a weighted average of current and past true returns [16]:

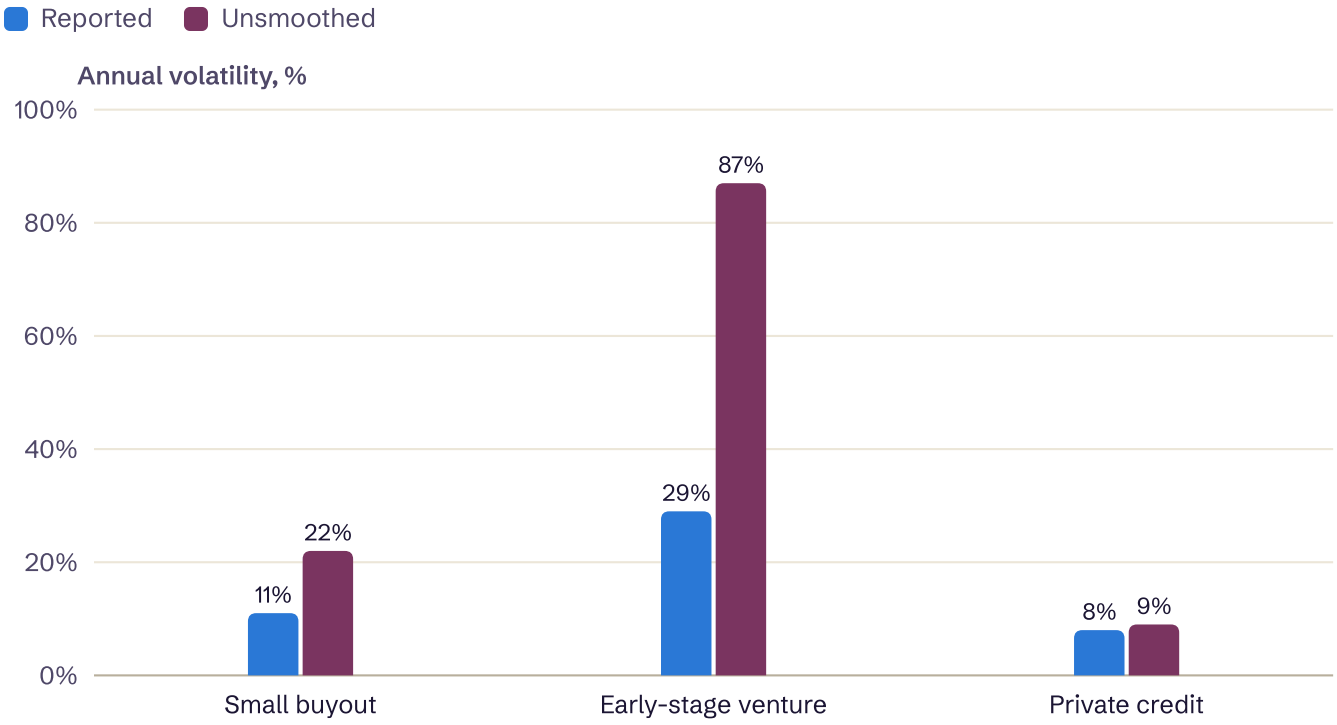
$$R^{\circ}_t = \theta_0 R_t + \theta_1 R_{t-1} + \ldots + \theta_k R_{t-k}, \text{ with } \Sigma\theta_j = 1, \text{ so } \sigma^{\circ 2} = (\Sigma\theta_j^2) \cdot \sigma^2$$

R°: reported return. R: true economic return. Because $\Sigma\theta_j^2$ is below one whenever the weights are spread over several quarters, reported volatility understates true volatility and reported Sharpe ratios overstate risk-adjusted performance.

Desmoothing inverts the filter. Regressing private returns on several lags of public returns, rather than only the current quarter, recovers most of the hidden beta. Anson did this for private capital from 1998 to 2022. Unsmoothed volatility doubled for small buyouts and tripled for early-stage venture, while private credit barely moved [17]. Figure 5 shows the gap.

Figure 5. Reported volatility understates risk, except in credit

Annual volatility of private capital returns, 1998–2022, as reported and after unsmoothing with lagged betas



Source: Anson (2024), as summarized by Alpha Architect and others [17].

Cliff Asness calls this volatility laundering [18]. His argument is that investors pay for the smoothing. An institution that reports a private portfolio with 10% volatility gets to hold more of it, rebalance less and explain fewer drawdowns to its board. If investors value that, they will accept a lower return for it, and the illiquidity premium can turn negative without anyone noticing.

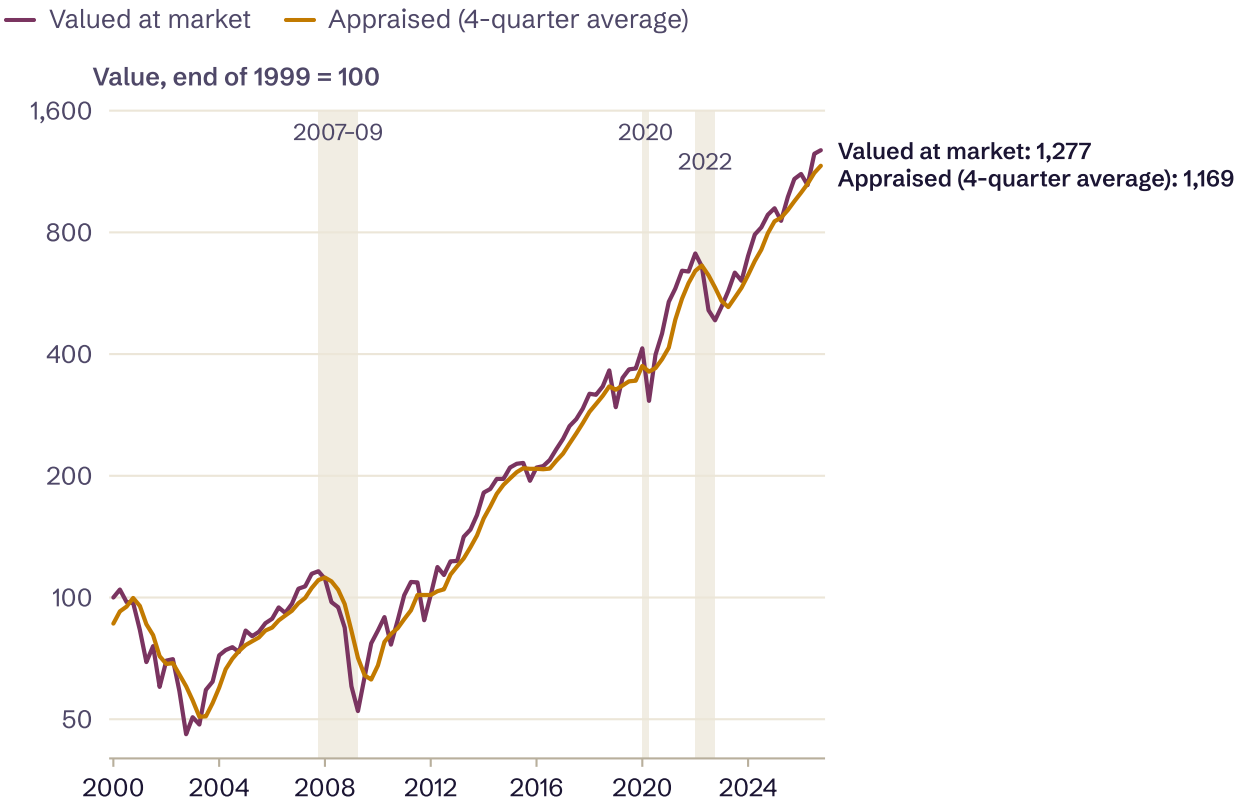
I think he is right about the mechanism and that its size depends on the investor. A pension whose board reads quarterly marks gets real value from smoothing. An individual who will need the cash in year four gets none, and is the one most likely to sell at a stale price that someone else profits from.

An experiment: mark the stock market like a private fund

The claim that smoothing manufactures low risk can be tested without any private data. I took the U.S. stock market from January 2000 to September 2026, levered it 1.3 times with Treasury bills to match the market beta of buyout investments [91, 93], and valued the result two ways at each quarter-end. The first is the market price. The second is an appraisal that averages the last four quarter-end values, which is the Getmansky–Lo–Makarov filter with equal weights [16]. Four quarters is not an arbitrary choice. Equal weights over four quarters halve measured volatility, and half is the ratio Anson finds between reported and unsmoothed returns for small buyouts [17]. Both versions hold the same stocks and pay the same cash.

Figure 6. Same portfolio, two sets of books

U.S. stock market levered 1.3 times, valued at market and as an average of the last four quarter-ends. End of 1999 = 100, log scale



My arithmetic. The portfolio earns 1.3 times the market's return over Treasury bills, plus the bill return, each month, 2000Q1 to 2026Q3. The appraised value at each quarter-end is the geometric average of the last four quarter-end market values. The data table shows every fourth quarter.

Source: Ken French data library, CRSP value-weighted U.S. market and one-month Treasury bills [93]; method from Getmansky, Lo and Makarov [16].

Figure 6 shows the two lines. They start and end in nearly the same place, because an average of past values has to catch up in the end. In between they tell different stories. In the first quarter of 2020 the portfolio lost 25.9% of its value and the appraised version lost 3.2%. In 2007–09 the true fall was 54.9% and the appraised fall 44.1%, spread over more quarters. In 2022 it was 31.7% against 21.1%.

Table 1. What the allocator sees

Quarterly statistics for the same levered stock portfolio under the two valuations, 2000Q1 to 2026Q3

Statistic	Valued at market	Appraised (4-quarter average)
Annual volatility	22.2%	11.7%
Beta to the U.S. stock market	1.30	0.32
Correlation with the stock market	1.00	0.46
Sharpe ratio	0.46	0.74
Worst quarter	–28.5%	–14.4%
Largest drawdown	–56.2%	–49.2%
Serial correlation of quarterly returns	0.01	0.73
Annualized return	10.0%	10.2%

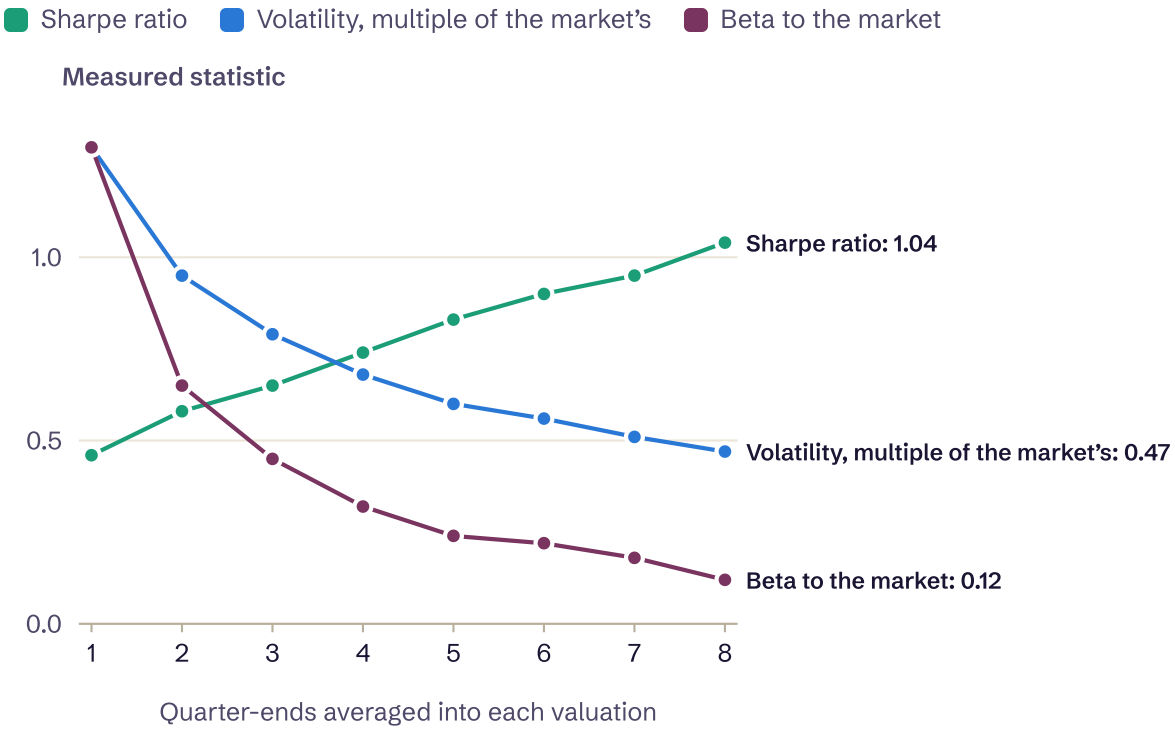
My arithmetic on 107 quarterly returns. Volatility is the standard deviation of quarterly returns times two. Beta and correlation are against the unlevered market in the same quarter. The Sharpe ratio uses the return over Treasury bills. Drawdowns are measured on quarter-end values.

Source: Ken French data library [93].

Table 1 has the statistics an allocator would be shown. Volatility falls from 22.2% to 11.7%. Beta falls from 1.30 to 0.32, so a levered stock portfolio looks like a diversifier. The Sharpe ratio rises from 0.46 to 0.74, the difference between an ordinary asset and an excellent one. None of it is real. The one statistic that gives it away is serial correlation: appraised returns predict themselves, at 0.73, and market returns do not, at 0.01. (The 0.2-point gap in annual return comes from where the two lines start and stop.)

Figure 7. More averaging, less measured risk

The same levered stock portfolio, by the number of quarter-ends averaged into each valuation



My arithmetic, 2000Q1 to 2026Q3. One quarter means valued at market. The unlevered market's volatility over the period was 17.1% a year.

Source: Ken French data library [93].

Figure 7 repeats the exercise for appraisal windows of one to eight quarters. Every extra quarter of averaging lowers measured beta and raises the measured Sharpe ratio. With two years of averaging, a 1.3-times levered stock portfolio reports a beta of 0.12 and a Sharpe ratio above 1. An optimizer given those inputs holds as much of the asset as its constraints allow. The allocation is produced by the calendar on which the asset is priced.

Real private equity looks the same when it is priced. Boyer, Nadauld, Vorkink and Weisbach built an index from the prices paid for fund stakes in the secondary market from 2006 to 2018. Its volatility is about 34%, against 14% to 20% for indexes built from NAVs [92].

Leverage, size and value

The last adjustment is risk. A buyout fund buys companies smaller than those in the S&P 500, often cheaper on earnings multiples, and adds debt. Compare it with the S&P 500 and you credit the manager with the premia for size, value and leverage.

Phalippou and Gottschalg, correcting for stale NAVs and sample bias, found that the average fund trailed the S&P 500 by 3% a year net of fees, and by 6% a year after adjusting for risk [19]. L'Her and coauthors built a bottom-up benchmark that levers a size- and sector-matched public index to each fund's portfolio, and found no significant outperformance on a dollar-weighted basis [20].

Stafford's replication is the most striking result [21]. A passive portfolio of small, low-EBITDA-multiple public stocks with modest leverage, valued at hold-to-maturity accounting instead of market prices, produced a return distribution very close to the pre-fee private equity index. The accounting alone changes the risk numbers: the same portfolio that shows a beta near 2 when marked to market shows a beta statistically indistinguishable from zero when marked like a private fund. The low risk lives in the bookkeeping.

Two studies put numbers on the risk. Franzoni, Nowak and Phalippou used the cash flows of 4,403 private equity investments and found a market beta of 1.3, a loading of 1.0 on the value factor and a liquidity beta of 0.64, higher than 86% of traded stocks. The liquidity risk premium alone was about 3% a year, and once it was counted the alpha before fees was zero [91]. Boyer and coauthors, pricing funds off secondary trades instead of cash flows, found a beta of 1.79 and an alpha of −2% a year, not statistically different from zero [92]. Private equity is paid for liquidity risk in the Pástor–Stambaugh sense, the same premium a public investor can collect. After that there is no return left over to attribute to the lockup.

4. The evidence, market by market

Table 2 summarizes what each market appears to pay for illiquidity, on the best public evidence I could find. The subsections show the work.

Table 2. What each market pays for illiquidity

Best public evidence, with my reading of it

Market	Evidence	Premium	My read
Treasuries	Refcorp vs. Treasury yields, 1991–2001 [8]; on-the-run vs. off-the-run 30-year bond [9]	10–16 bp; 12 bp on one day in 2001	Tiny in normal times, large in a flight to safety
Corporate bonds	Non-default share of the spread, CDS-implied [10]	49% of the spread (AAA/AA) to 17% (BB)	Collectible daily, with no lockup
U.S. stocks: illiquidity level	Characteristic premium [7]	About 1.8%/yr in the 1960s, near zero since outside microcaps	Mostly gone
U.S. stocks: liquidity risk	High minus low liquidity beta, 1966–1999 [5]; replications [6]	7.5%/yr; larger since 1999	The premium that lasts
Buyouts	Cambridge U.S. PE index vs. S&P 500 mPME [24]	+2.4 pts/yr over 25 years; +0.1 over 10	Negative once leverage is counted [91, 92]
Venture	Top vs. bottom quartile IRR [30]; outcome skew [28, 29]	About 20 pts quartile spread	Access matters more than the asset class
Direct lending	Cliffwater index vs. leveraged loans, 2025 [32, 34]; buyout loan spread vs. syndicated [35]	3.4 pts total in 2025; about 1.5 pts like-for-like	Real in the contract; used up by fees and smoothing in AQR's estimate [26]
Core real estate	NPI vs. REITs, 2021–2025 (Figure 11)	Five-year totals of about +21% vs. +27% (my arithmetic)	No visible premium, only a delay
Infrastructure	Closed-end fund cash flows and returns [49]	Weak risk-adjusted performance	Unproven
Secondaries	LP stake prices, 2025 [54]	Buyout 92%, venture 78%, real estate 70% of NAV	About 2%/yr for buyout, 5%/yr for venture (implied, Section 5). This is the side that gets paid

Source: As cited in each row; see Sections 2 through 5.

Public bonds and equities

The cleanest test of a pure liquidity premium is two bonds with the same cash flows and the same credit that differ only in how easily they trade. Longstaff compared Treasuries with bonds issued by Refcorp, which carry the same full government guarantee, and found Refcorp yields 10 to 16 basis points higher from 1991 to 2001. The liquidity premium was worth more than 15% of the value of some Treasury bonds, and it widened when investors fled to safety [8]. Krishnamurthy showed the newest 30-year Treasury yielding 12 basis points less than the previous issue, a bond with almost the same cash flows, on a single day in February 2001 [9].

That is a small premium in normal times and a large one in a panic. March 2020 made the point at scale. Dealers could not absorb the selling, bid-ask spreads in the deepest market in the world blew out, and the Federal Reserve bought nearly $2 trillion of Treasury notes and bonds in 2020 to restore it [12].

In corporate bonds, Longstaff, Mithal and Neis used credit default swaps to split spreads into default and everything else. Default explained 51% of the spread on AAA and AA bonds, 56% on A, 71% on BBB and 83% on BB [10]. The non-default part, which moves with bond-level and market-wide illiquidity, is about half the spread on the safest bonds. Chen, Lesmond and Wei found that liquidity explains up to half of the cross-sectional variation in spread levels [11]. An investor who wants a liquidity premium with daily marks and no lockup can collect one in high-grade corporate bonds.

In stocks, the premium for illiquidity as a characteristic has faded. Ben-Rephael, Kadan and Wohl found it fell from about 1.8% a year in the 1960s and early 1970s to an amount indistinguishable from zero, except among the smallest stocks [7]. The premium for liquidity risk has not faded [6]. Public markets are telling you which kind of illiquidity gets paid: the kind that hurts in a crash.

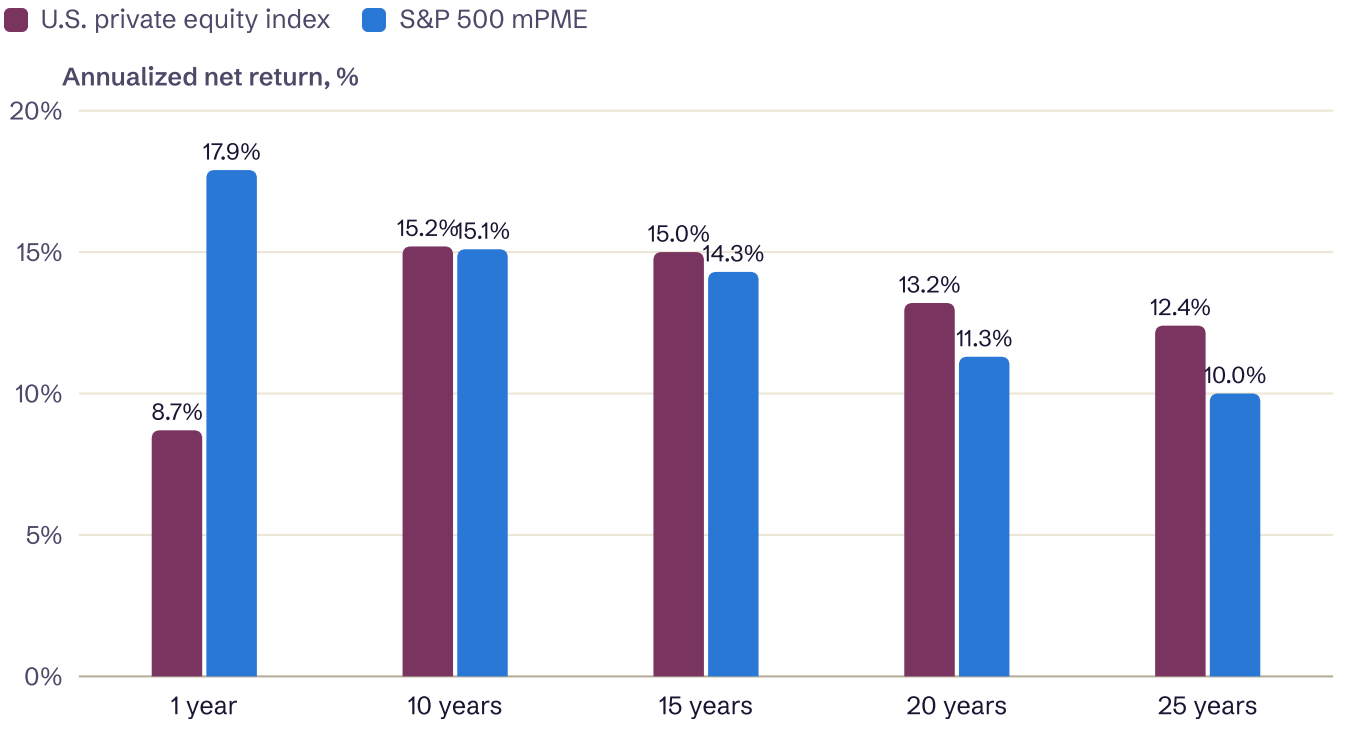
Buyouts

The best academic evidence on buyouts is more favorable than the replication work above, which is why the debate has lasted. Harris, Jenkinson and Kaplan, using cash flows reported by institutional investors, found that buyout funds beat the S&P 500 by 20% to 27% over a fund's life, or more than 3% a year [22]. A later study by the same authors with Stucke put the average buyout PME at 1.20, falling to 1.11 for vintages after 2005 [23]. Kaplan and Schoar's earlier sample found average net returns about equal to the S&P 500 [13].

The most current public benchmark tells the same story through time. Cambridge Associates' U.S. private equity index, which covers 1,769 buyout and growth equity funds, returned 12.4% a year over the 25 years to December 2025, against 10.0% for the same cash flows invested in the S&P 500. Over the last ten years the gap was 0.1 points, and public markets won most horizons shorter than ten years [24]. Figure 8 shows each horizon.

Figure 8. The buyout edge is a long-horizon number

Cambridge Associates U.S. private equity index vs. the same cash flows in the S&P 500 (mPME), annualized, periods ending December 31, 2025



Index returns are pooled horizon IRRs net of fees, expenses and carry. The mPME invests and withdraws the same cash flows in the S&P 500.

Source: Cambridge Associates, US PE/VC Benchmark Commentary: Calendar Year 2025 [24].

Three forces explain the shrinking gap. Purchase prices rose as capital crowded into the strategy. Debt got more expensive after 2022, which removed the cheapest source of buyout returns. And exits slowed, which leaves more of recent performance as unrealized marks [62, 63]. Cambridge also notes that private equity beat the Russell 2000 more often than it beat the S&P 500 [24], which is what you would expect if much of the historical edge came from owning smaller companies with debt.

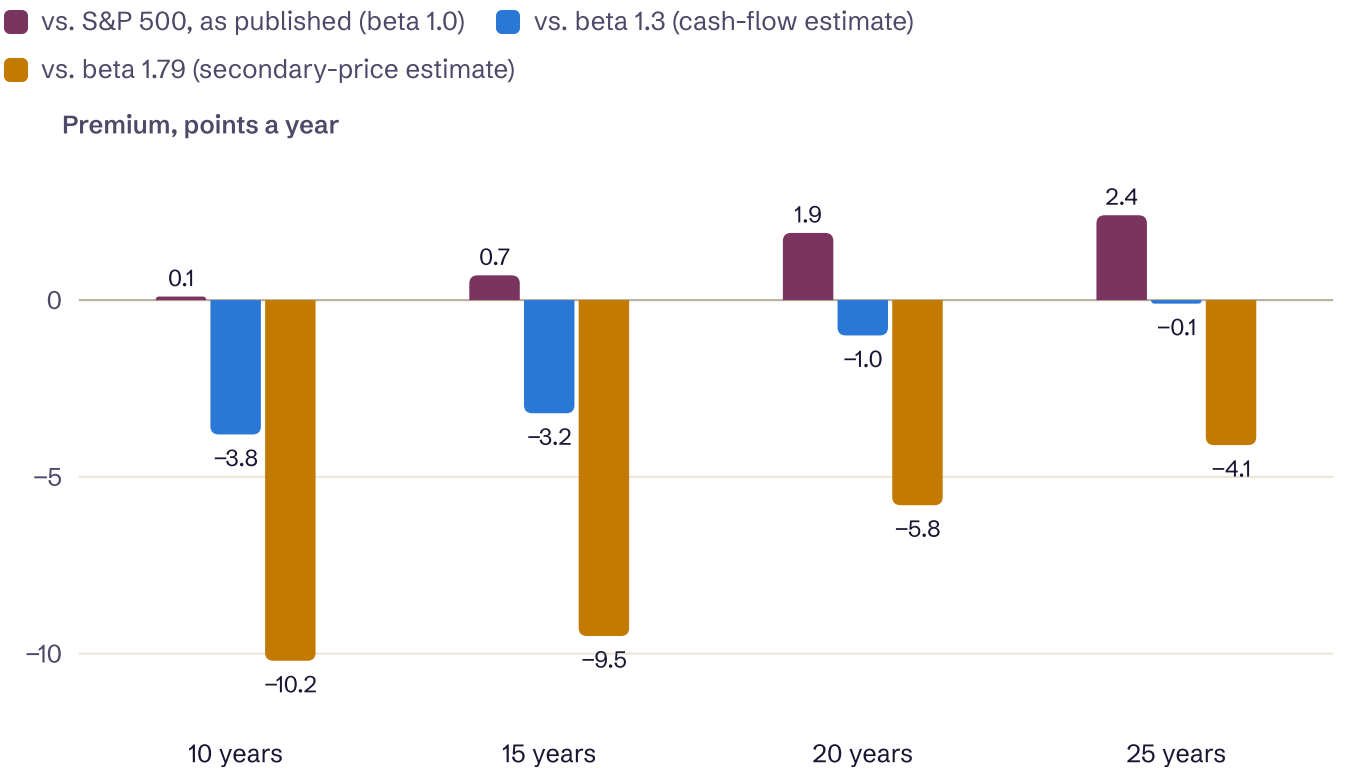
The Cambridge comparison holds the benchmark at a beta of one. Buyouts are not a beta-one asset: AQR puts debt at 69% of equity in the average deal [26], and the studies in Section 3 put the market beta between 1.3 and 1.79 [91, 92]. If a fund has a beta of β, the public portfolio with the same market risk holds β in the index, financed at the Treasury bill rate, and the fund's excess over that portfolio is:

$$\alpha \approx (R_{PE} - R_m) - (\beta - 1)(R_m - r_f)$$

R_{PE}: the Cambridge index. R_m: the S&P 500 on the same cash flows. r_f: the average Treasury bill return over the horizon, which was 2.11% a year for the ten years to December 2025, 1.41% for 15, 1.61% for 20 and 1.72% for 25 [93]. An approximation, because it applies a one-period formula to annualized returns.

Figure 9. The buyout premium, before and after matching leverage

Cambridge Associates U.S. private equity index less a public portfolio with the same market beta, points a year, periods ending December 31, 2025



My arithmetic: premium = (PE − S&P 500 mPME) − (β − 1) × (S&P 500 mPME − Treasury bill return). An approximation, since it applies a one-period formula to annualized dollar-weighted returns.

Source: Cambridge Associates [24]; betas from Franzoni, Nowak and Phalippou [91] and Boyer et al. [92]; Treasury bill returns from the Ken French data library [93].

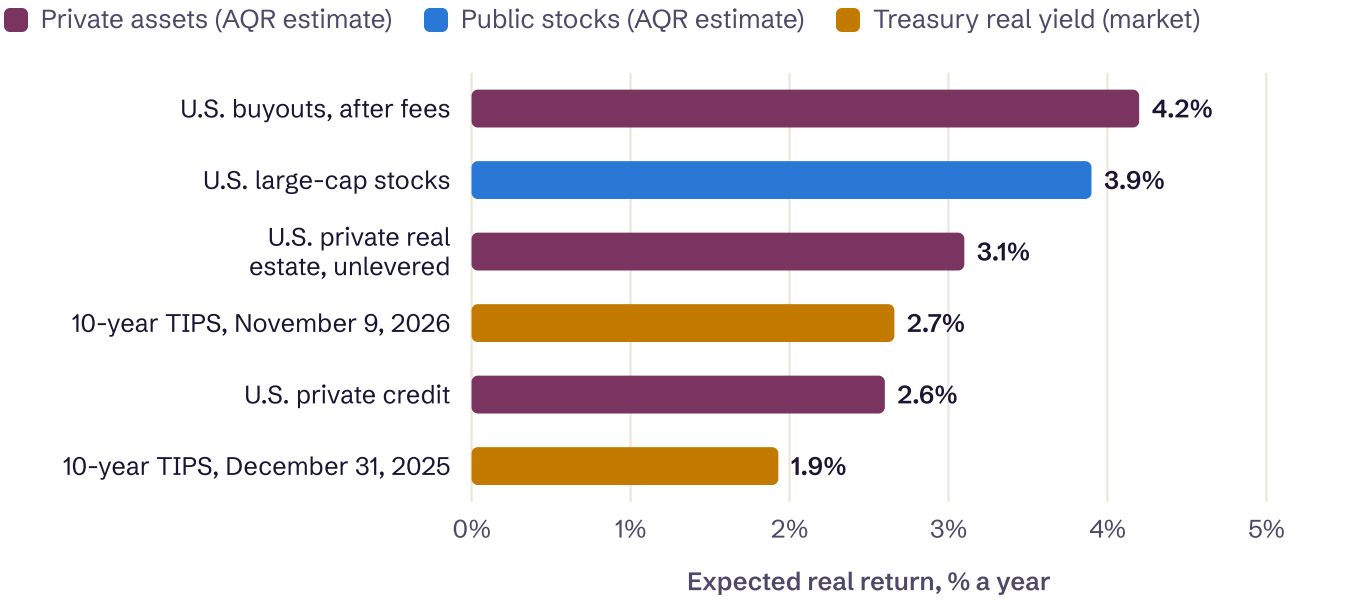
At a beta of 1.3 the premium is −0.1 points a year over 25 years, −1.0 over 20, −3.2 over 15 and −3.8 over ten. At 1.79 it runs from −4.1 to −10.2. The published 2.4 points is almost exactly what 30% more market exposure earned over that quarter century (my arithmetic).

The usual reply is that buyouts should be compared with small companies. That was a fair benchmark for the buyouts of the 1990s. It is not what the funds own now: information technology is 36% of the Cambridge index by value, and managers put 37% of the capital they invested in 2025 into it [24]. A portfolio that is a third software, with leverage, is not a small-cap value portfolio.

Forward-looking estimates point the same way. AQR's 2026 capital market assumptions put the expected real return on U.S. buyouts at 4.2% a year after fees, against 3.9% for U.S. large-cap stocks [26]. That is 0.3 points for more leverage and a ten-year lockup. The same document expects 2.6% from private credit and 3.1% from unlevered private real estate [26]. A ten-year Treasury inflation-protected bond, which can be sold in seconds, yielded 1.93% real when AQR made those estimates and 2.66% on November 9, 2026 [98].

Figure 10. What you are expected to be paid

Expected real return over the next five to ten years, % a year, with the real yield on a ten-year Treasury for scale



AQR's estimates are as of December 31, 2025 (buyouts and real estate, September 30, 2025). Treasury yields are par real yields: 1.93% on December 31, 2025 and 2.66% on November 9, 2026. The yields are not forecasts; they are what the bond pays if held to maturity.

Source: AQR, 2026 Capital Market Assumptions [26]; U.S. Treasury [98].

Ilmanen, Chandra and McQuinn concluded that private equity no longer offers the net-of-fee edge it did 15 to 20 years ago, looking either backward or forward [25]. Six years later their firm's own numbers have the edge at a third of a point.

Venture capital

Venture is where averages mislead most. Correlation Ventures looked at more than 21,000 financings from 2004 to 2013 and found that 65% returned less than the capital invested and 4% returned ten times or more [28]. Horsley Bridge's data on about 7,000 investments by funds it backed from 1985 to 2014 showed that the 6% of deals returning at least ten times produced 60% of all returns [29]. With payoffs that skewed, the asset-class average tells you little about what any one LP earns.

Fund-level data agree. Harris, Jenkinson and Kaplan found that venture funds beat public equities in the 1990s and trailed them in the 2000s [22]. PitchBook puts the gap between top- and bottom-quartile venture funds at close to 20 percentage points of IRR [30]. Venture had a strong 2025, when the Cambridge U.S. venture index returned 21.1% [24], but PitchBook notes that the AI boom drove it, with gains concentrated in the largest funds and a few blockbuster IPOs [31].

Venture is also where performance persistence has held up. For buyouts, Harris, Jenkinson, Kaplan and Stucke find little persistence after 2000 using only what an investor knows at fundraising. For venture it persists, though more weakly than before [23]. In venture, access is the premium. A venture allocation without the top managers is a bet on the median of a power law, and the median of a power law is poor.

Private credit

Private credit gives the most measurable premium because loans pay contractual coupons. The Cliffwater Direct Lending Index returned 9.3% in 2025: 10.4% of interest income, less about 1.1 points of losses and markdowns (my arithmetic) [32]. The Morningstar LSTA U.S. Leveraged Loan Index, which holds broadly syndicated loans to larger companies, returned 5.90% [34]. Over twenty years the direct lending index has averaged roughly 9.5% a year, with realized credit losses of about 1% a year [32, 33].

Not all of the 3.4-point gap in 2025 is an illiquidity premium. Direct loans go to smaller companies, often with more leverage, and the spread pays for both. On comparable buyout loans, Morgan Stanley Investment Management cites an average spread of 153 basis points over the syndicated market [35]. That is my best estimate of the pure illiquidity premium in senior corporate lending: about 1.5 points a year before fees, which is attractive precisely because loan funds charge less than equity funds.

The credit cycle is now testing it. Fitch's default rate for the private credit borrowers it rates privately reached a record 9.2% in 2025, up from 8.1% in 2024 [36]. The IMF had already flagged a market of about $2 trillion for fragile borrowers, layered leverage, semi-liquid vehicles and stale valuations [37]. The BIS has examined how much of BDC lending goes to software companies exposed to AI disruption [38], and J.P. Morgan Asset Management puts that exposure at about 20% of BDC portfolios [39]. After the failures of Tricolor and First Brands in the fall of 2025, Jamie Dimon warned that "when you see one cockroach, there are probably more" [40].

The public market has an opinion on these loans. About 50 publicly traded BDCs hold the same kind of loans as the non-traded funds, often at the same managers, and in February 2026 they traded at an average discount of more than 20% to NAV [94]. Octus puts the discount near 20%, with some large funds near 50%, and finds average recoveries in private credit restructurings of about 50 cents on the dollar, where managers have cited 70 [101]. A non-traded fund sells and redeems its shares at NAV [94]. Either the public price is wrong or the NAV is.

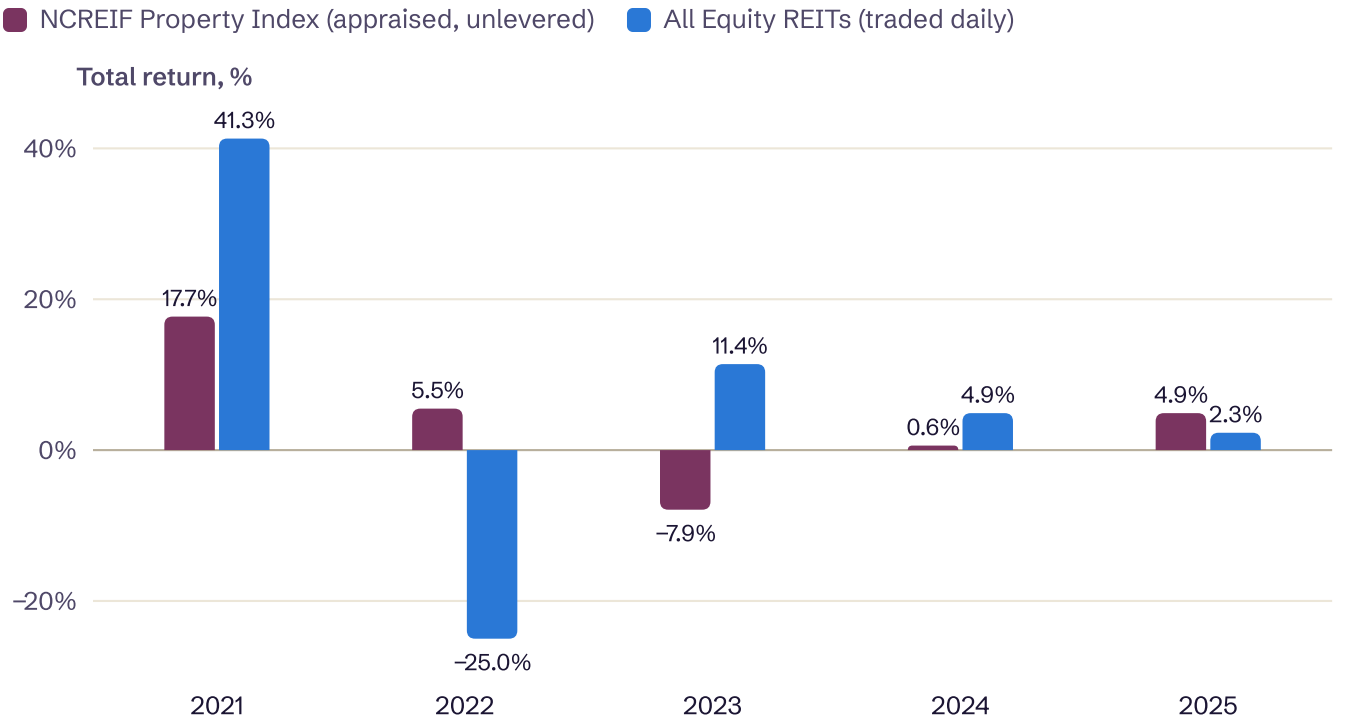
The contractual spread is real. What the evidence does not support is the idea that it comes with low risk. Private credit's reported volatility, 8% reported and 9% desmoothed in Anson's data [17], describes the last cycle and says nothing about a default cycle. AQR's estimate assumes that fees and the value investors place on smooth marks use up whatever premium there is [26].

Private real estate

Real estate is the best natural experiment on smoothing, because similar buildings trade in two wrappers. REITs trade every second. Private core funds and the NCREIF Property Index (NPI) are appraised. Figure 11 shows calendar-year returns for both from 2021 to 2025.

Figure 11. Same buildings, different marks: appraised vs. traded real estate

Calendar-year total returns, NCREIF Property Index vs. FTSE Nareit All Equity REITs



NPI 2022 is my compounding of NCREIF's quarterly returns (5.33%, 3.23%, 0.6%, −3.5%). NPI 2024 is the Expanded NPI; NPI 2025 is the trailing four quarters. REIT 2022 return was −24.95%; 2025 is Nareit's year-to-date figure at December 31.

Source: NCREIF releases and coverage [41-46]; Nareit [47, 48].

In 2022 REITs lost 25% while the NPI gained about 5.5%. The NPI did not post a losing quarter until the fourth, at −3.5% [41, 42, 48]. In 2023 the order reversed: REITs gained 11.4% while appraisals finally caught down and the NPI lost 7.9% [43, 48]. Over the five years the two compound to similar places, about +21% for the unlevered NPI and about +27% for REITs, by my arithmetic from the annual figures.

The private wrapper did not avoid the 2022 repricing. It booked it a year late. Anyone who redeemed from a private real estate fund at 2022 appraisals was paid out at stale, high values, at the expense of the investors who stayed. I find no premium for illiquidity in core real estate over this cycle, only a delay.

Infrastructure

Infrastructure is sold on stable, inflation-linked cash yields. Fund data do not show that. Andonov, Kräussl and Rauh found that the cash flows of closed-end infrastructure funds are about as volatile and cyclical as those of other private equity, that returns depend on quick exits in the same way, and that risk-adjusted performance is weak [49]. Public pension funds, the biggest buyers, did worse than private investors, and the authors attribute 25% to 40% of their higher allocations and about 30% of their underperformance to ESG preferences and regulation [49].

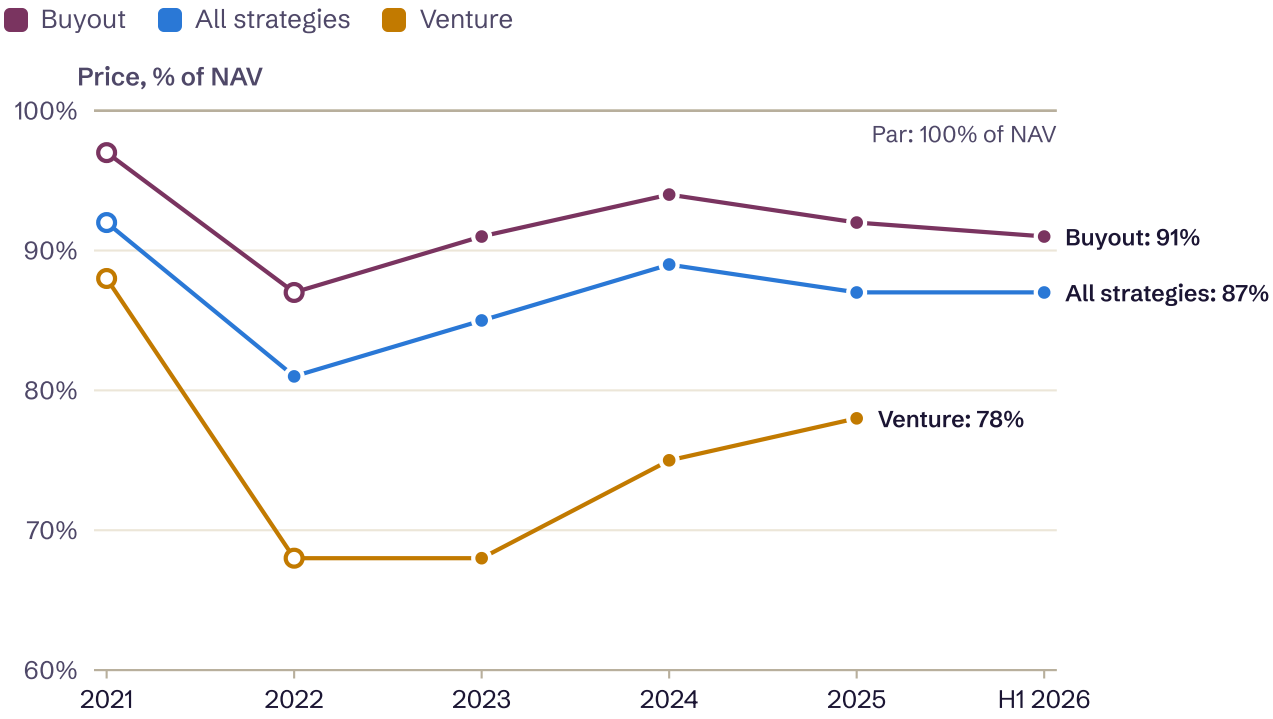
The illiquidity in infrastructure is real. I have not seen public evidence of a premium for it beyond what listed infrastructure plus leverage would provide.

5. Secondaries: the market price of illiquidity

Everything above infers the premium from returns. The secondary market prices it directly. When an LP sells a fund stake, the buyer's price is quoted as a percentage of the latest NAV. The discount is what the seller gives up for cash today, plus any disagreement about whether the NAV is right. No other number in private markets is closer to a quote for illiquidity.

Figure 12. The price of liquidity: what buyers pay for LP fund stakes

Average LP portfolio pricing in the secondary market, % of NAV



Hollow markers are implied from changes Jefferies reports rather than stated levels: 2021 from the declines to mid-2022 and to full-year 2022 [50, 51]; 2022 buyout from the 400 bp rise to 91% in 2023; 2022 venture from the statement that 2023 pricing matched the prior year [52]. H1 2026 all-strategy pricing is "approximately 87%." Venture for H1 2026 is not shown.

Source: Jefferies Global Secondary Market Reviews, 2022-2026 [50-55].

Figure 12 shows Jefferies' average prices for LP portfolios. Buyout stakes have sold within a few points of NAV in every year since 2023: 91% in 2023, 94% in 2024, 92% in 2025 and 91% in the first half of 2026 [52-55]. Venture has been far cheaper, at 68% in 2023, 75% in 2024 and 78% in 2025 [52-54]. Real estate fell to 70% in 2025 [54]. The 2022 line shows how fast pricing moves when public markets fall: the average across strategies dropped 11 points in a year [51].

These discounts convert into an implied premium. If a buyout stake with about four years of remaining life sells at 92% of a fair NAV, the buyer earns $(1/0.92)^{1/4} - 1$, about 2.1% a year, on top of the fund's own return. The same arithmetic gives about 5.1% a year for venture at 78% over five years. These are rough, because they assume the NAV is right and the remaining life is what I guessed. But they land close to the theory. The market charges about

2 points a year to take buyout illiquidity off a seller's hands, and more than twice that for venture, where NAVs are trusted less and remaining lives are longer.

Figure 13. What a discount is worth to the buyer

Extra annual return from buying a fund stake below NAV, by price paid and years until the fund pays out

Price, % of NAV	Years until the fund pays out				
	2	3	4	5	6
54% (buyout, 2009)	36.1%	22.8%	16.7%	13.1%	10.8%
70% (real estate, 2025)	19.5%	12.6%	9.3%	7.4%	6.1%
78% (venture, 2025)	13.2%	8.6%	6.4%	5.1%	4.2%
87% (all strategies, H1 2026)	7.2%	4.8%	3.5%	2.8%	2.3%
91% (buyout, H1 2026)	4.8%	3.2%	2.4%	1.9%	1.6%
94% (buyout, 2024)	3.1%	2.1%	1.6%	1.2%	1.0%

My arithmetic: $(1 \div \text{price})^{1/T} - 1$. It assumes the NAV is right and is received T years later, so it overstates the gain when marks are stale. Outlined: buyout stakes over four years, in 2009 and in the first half of 2026.

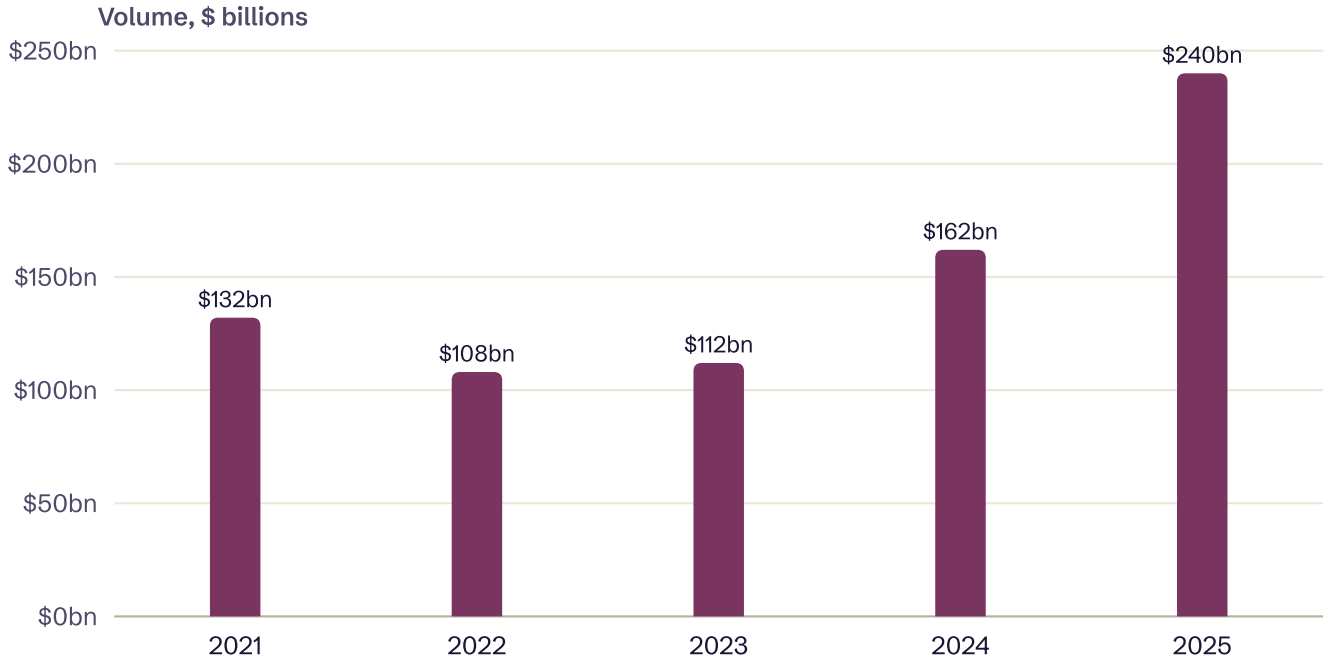
Source: Prices from Jefferies [53-55] and Nadauld et al. [59].

Figure 13 does the conversion for every price and remaining life in this section. The numbers in the top row are what a liquidity supplier earns in a crisis. The numbers in the bottom rows are what the market charges in a calm year, and they are the closest thing there is to a live quote for the illiquidity premium on a seasoned buyout fund: about 2.4% a year at 91% of NAV with four years left, and 1.6% with six.

In a crisis the quote collapses. Nadauld, Sensoy, Vorkink and Weisbach studied secondary transactions from 2006 to 2014 and found an average discount of 13.8% of NAV, closer to 9% for the typical four- to nine-year-old fund, and larger discounts when the economy was weak [59]. In their data buyout stakes changed hands at about 54% of NAV in 2009. Over four years that discount is worth roughly 17% a year to the buyer on top of the fund's return. That is the liquidity premium in its purest form, paid by forced sellers to whoever has cash.

Figure 14. A real market for liquidity

Global secondary transaction volume, LP-led and GP-led combined, $ billions



2023 is Jefferies' approximate figure ("around $112 billion"). The first half of 2026 alone was $118 billion [55]. Lazard's estimate for 2025 is $233 billion [57].

Source: Jefferies [51–54, 56].

The market has grown into a real source of liquidity. Jefferies counted $132 billion of volume in 2021, $108 billion in 2022, about $112 billion in 2023 and $162 billion in 2024 [51–53, 56], then a record $240 billion in 2025 [54]. Lazard's independent estimate for 2025 is $233 billion, split almost evenly between LP-led and GP-led deals [57]. The first half of 2026 set another record, $118 billion by Jefferies' count, with GP-led deals at $62 billion or 53% of the total [55]. Dedicated secondary capital reached a record $327 billion in 2025 [54].

Most of the growth is GP-led. In a continuation vehicle the manager moves one or more companies from an old fund into a new vehicle it also manages, and existing LPs choose to cash out or roll. Continuation vehicles made up 89% of GP-led volume in the first half of 2026 [55]. Evercore found that most single-asset deals priced at NAV and about 14% priced above it [58].

A price at or above NAV on an asset the GP chose to keep is not evidence of a small liquidity discount. It is evidence that the GP picked its best asset. The LP deciding whether to roll should know that the person setting the price is on both sides of the trade.

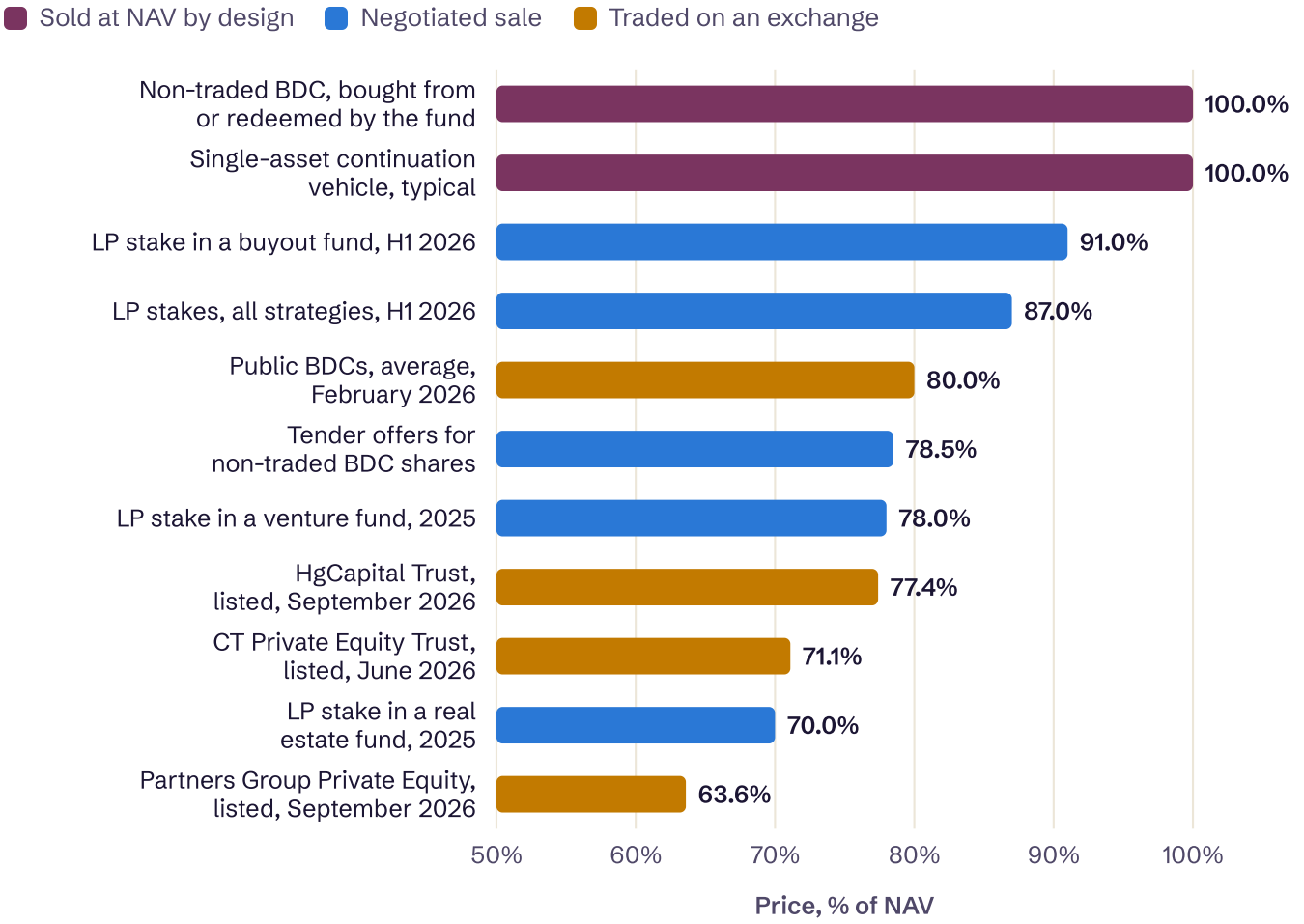
Even the most patient investors now use this market. In 2025 Yale was reported to be close to selling about $2.5 billion of private equity and venture stakes at a discount of less than 10% to NAV [60]. A decade ago a sale like that would have read as distress. Today it is portfolio management, and the price shows how little buyers ask for taking quality buyout exposure off a seller's hands in a calm market.

One asset, five prices

Put the quotes side by side and the argument gets simpler. Figure 15 lists what a dollar of reported NAV in a private fund sold for in 2025 and 2026, by the door it left through.

Figure 15. One kind of asset, many prices

What $100 of reported NAV in a private fund sold for in 2025-26, by how it was sold



Public BDCs: Raymond James' average discount was "more than 20%," so 80% is an upper limit. Tender offers: midpoint of offers 13% to 30% below NAV. Listed trusts are 100% less the reported discount (22.6%, 28.9% and 36.4%). Continuation vehicles: most single-asset deals priced at NAV, and about 14% above it.

Source: Barron's [94]; Evercore [58]; Jefferies [54, 55]; Cox Capital and Ares filings [74]; interactive investor [96]; Scottish Financial News [95].

The assets behind these prices are not identical, but they are the same kind: diversified private companies and private loans, valued by the same appraisal process. The spread from top to bottom is 36 points. The investors who transact at 100 are the ones entering non-traded funds and rolling into continuation vehicles. The investors who transact at 64 to 91 are buying from someone who needs out, in a negotiated sale or on an exchange. That gap is the illiquidity premium, and the chart shows who pays it and who receives it.

One version of this trade is visible in fund accounting. A fund that buys a stake at, say, 85% of NAV can carry it at NAV at the next valuation date and book the difference as a gain. The gain is real only if the NAV is, and

a fund that depends on it needs a steady supply of new discounted purchases, which means a steady supply of new money.

6. When liquidity disappears

Liquidity is not a fixed property of an asset. It is a property of an asset in a particular market, and it disappears in the states of the world where investors need it. Every private-markets stress episode has the same three parts: public portfolios fall, distributions stop, and capital calls keep coming.

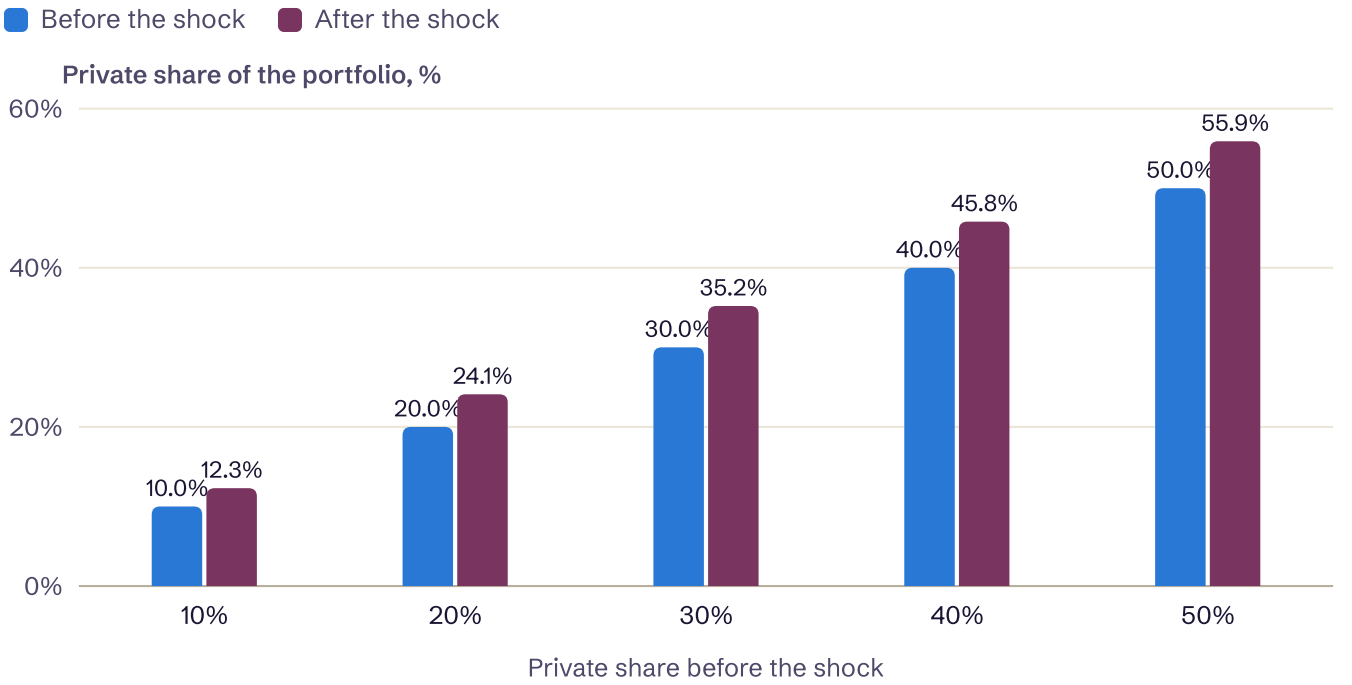
2008–09: the denominator effect

Harvard's endowment lost 27.3% in fiscal 2009, falling from $36.9 billion to $26.0 billion, and Yale's lost 24.6% [61]. Harvard, which also carried large uncalled commitments, cut them by about $3 billion [61], in a secondary market where buyout stakes traded at about half of NAV [59].

The mechanism is arithmetic. Take a portfolio that is 70% public and 30% private. Public equities fall 25%. Private marks, which lag, fall 5%. The private share rises from 30% to 35% without the investor doing anything, and with capital calls still arriving it keeps rising. The investor is now overweight the asset it cannot sell and has to sell what it can, near the bottom. (This example is mine.)

Figure 16. The denominator effect

Share of the portfolio in private assets before and after public assets fall 25% and private marks fall 5%



My arithmetic: w × 0.95 ÷ (w × 0.95 + (1 − w) × 0.75). No capital calls, distributions or spending, all of which push the share higher in a real drawdown.

Source: My illustration; the shock is the one described in the text.

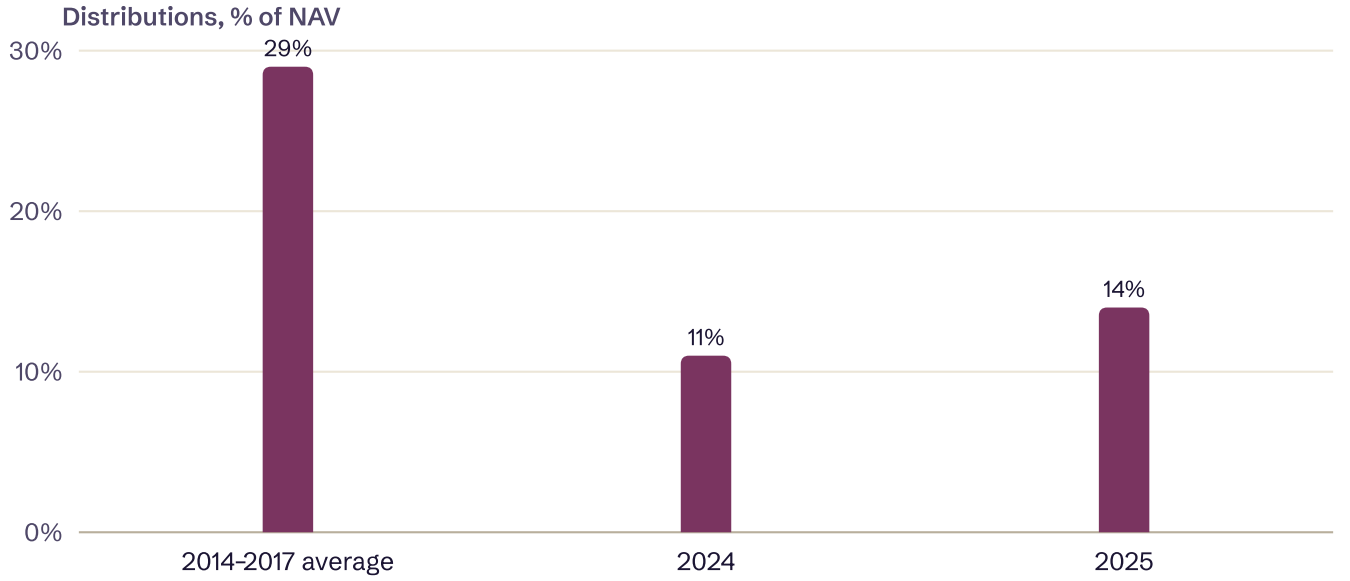
Figure 16 runs the same shock from other starting points. An investor at 20% private ends at 24%, one at 40% ends at 46%, and one at 50% ends at 56%, before a single capital call. The overshoot grows with the allocation, so the investors most exposed to a freeze are the ones the freeze pushes furthest past their limits.

2022-25: distributions dry up

2022 repeated the denominator effect on a smaller scale, and the aftermath has lasted longer. Exits slowed as rates rose, and buyout funds stopped returning cash. Bain reports that distributions averaged 29% of NAV from 2014 to 2017 and fell to 11% in 2024, the lowest in more than a decade [62]. They recovered only to about 14% in 2025, the fourth straight year below 15%, an industry record [63]. Bain's mid-2026 update puts the implied capital cycle at about seven years [64]. About 32,000 companies worth $3.8 trillion are unsold, and almost 40% of them have been held for more than five years, up from 29% in 2019 [63].

Figure 17. Distributions dried up

Global buyout fund distributions as a share of NAV



2025 is Bain's approximate figure ("about 14%"), the fourth straight year below 15%.

Source: Bain & Company, Global Private Equity Report 2025 and 2026 [62, 63].

For an LP whose pacing model assumed distribution yields of 25% to 30%, this is the liquidity crisis, even without a market crash. Commitments made in 2021 against expected distributions are being funded from other assets. That is why secondary volume set records in 2024 and 2025 while private equity returns were ordinary: sellers were buying liquidity, not escaping losses.

Semi-liquid funds and their gates

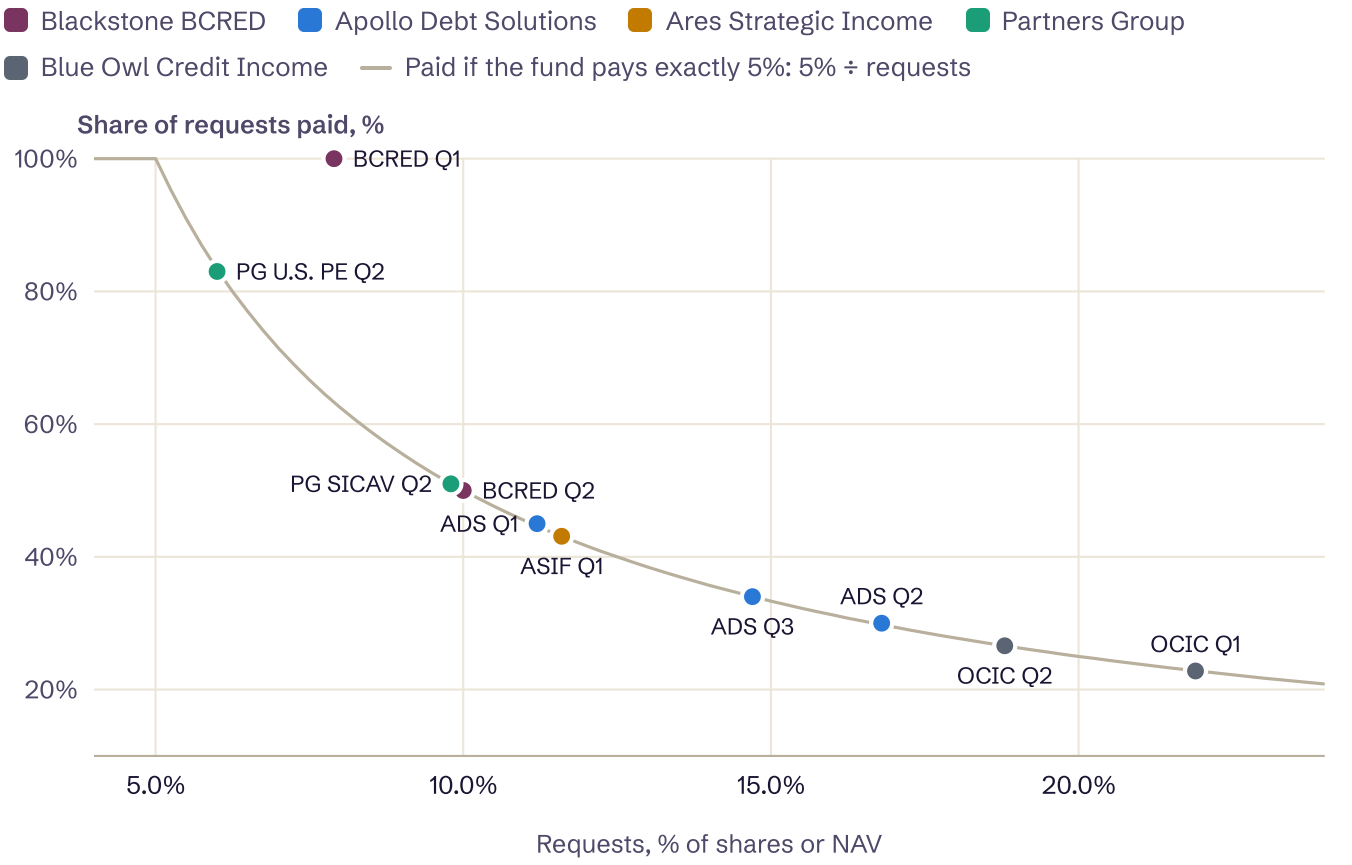
Evergreen and semi-liquid funds promise periodic liquidity, usually up to 5% of NAV a quarter, while holding assets that cannot meet a run. Their gates are not a malfunction. They are how the product works under stress, and the stress has now arrived twice.

The first test was BREIT. In November 2022 redemption requests exceeded its limits of 2% of NAV a month and 5% a quarter, and it paid 43% of each request [65]. Requests peaked at $5.3 billion in January 2023 [66]. The fund prorated for 15 months, returned more than $15 billion over that stretch, and paid all requests again in February 2024 [66]. In between, the University of California invested $4 billion, with Blackstone pledging $1 billion of its own BREIT shares to backstop an 11.25% annualized net return for UC over six years [67]. An investor willing to supply liquidity to a gated fund was paid well for it, on terms only an institution could negotiate.

The second test is this year, and it is in private credit. Figure 18 plots quarterly redemption requests at large semi-liquid funds in 2026 against the share of requested money investors actually received.

Figure 18. A 5% gate in practice: semi-liquid funds in 2026

Quarterly redemption requests vs. the share of requested money paid out



Shares paid are stated by the funds for BCRED Q1 (100%), ADS Q1 (about 45%), ADS Q3 (about 34%) and ASIF Q1 (43.1%). For the others I divide the 5% paid by the reported request. ASIF's request (about 11.6%) is implied by 5% being 43.1% of requests. Partners Group requests are approximate (about 9.8% and 6%). Blue Owl Credit Income (OCIC) is 5% divided by its reported requests. Blue Owl's technology fund, at 40.7% and 38.1%, is off the chart to the right.

Source: Fund filings and notices [68-71]; Bloomberg via Swissinfo [72]; Reuters [97].

Requests at Apollo Debt Solutions reached 11.2% of shares in the first quarter, 16.8% in the second and 14.7% in the third, and the fund paid 5% each time [68, 69]. Ares Strategic Income Fund paid 5% of shares in the first quarter, which was 43.1% of what holders asked for [70]. Blackstone's BCRED met all of a 7.9% request in the first quarter by raising its cap to 7% while Blackstone and its employees bought shares, then paid 5% against requests

of about 10% in the second [71]. Partners Group capped its Luxembourg evergreen fund at 5% when requests approached 10% of NAV, and its U.S. private equity master fund when they reached about 6% [72]. Blue Owl's largest credit fund received requests for 21.9% of shares in the first quarter and 18.8% in the second and paid 5% each time. At its technology lending fund the requests were 40.7% and 38.1% [97]. Fitch expects requests to stay above 5% through 2026, and judges that its rated BDCs can keep paying 5% a quarter for a year without new money coming in [73].

Every point in Figure 18 sits on or near the curve 5% divided by requests. That curve is the whole product. A quarterly liquidity promise is a promise of 5% of NAV, shared among everyone who asks. When one shareholder in eight asks, each gets about 40 cents on the dollar, and the rest waits in a queue that lengthens as unfilled requests are resubmitted. At Blue Owl's technology fund, where four in ten asked, each got about 12 cents (my arithmetic).

The price of skipping the queue is now quoted too. Cox Capital has made tender offers for shares of several non-traded BDCs at 13% to 30% below their reported NAVs [74]. Ares told shareholders that none of its Class I holders took an earlier offer [74]. Both facts are informative. Most holders would rather wait than pay 15% for cash today, and the ones who can't wait now have a price.

NAV loans and continuation vehicles

Managers have built tools to manufacture liquidity without selling. NAV loans borrow against a fund's portfolio, often to pay distributions. Oaktree, majority owner of the NAV lender 17Capital, sized that market at about $44 billion in 2023 and more than $70 billion in 2025 [75]. Continuation vehicles extend holding periods while offering LPs an exit [55, 58].

Both can help an LP who needs cash. Both also turn an illiquidity problem into something else: leverage on the LP's own equity in the case of NAV loans, and a conflicted price in the case of continuation vehicles. A distribution funded by a NAV loan is not DPI in any sense that matters. It is a loan against the investor's own assets, with interest.

7. Who should own illiquidity, and how much

Theory is clear about who should hold illiquid assets: investors whose horizon is longer than the lockup and whose need for cash is low and predictable. That describes a perpetual endowment better than a pension, and a pension better than an individual. The industry's growth is running in the other direction.

How much is a separate question from who. The model in Section 2 gives the scale: about a twelfth of the weight an asset would get if it were liquid, for something that trades once a decade and pays no premium [99]. I work the sizing through for individual investors, by age and risk tolerance, in a companion paper, The Brockmeier Model. There the default weight on private funds is zero, and anything above zero has to fit inside a liquidity budget.

Endowments and pensions

U.S. college endowments held about 54% of their assets in private and alternative strategies in fiscal 2025, including real assets, and returned 10.9% on average [76, 78]. Allocations scale with size: institutions under $50

million held 12.5% in alternatives, while those over $5 billion held 62.5% [77]. The large endowments are the investors Amihud and Mendelson's clientele argument describes.

They also show its limits. Yale, the model's originator, returned 11.1% in fiscal 2025 and trailed public indexes, with its CIO pointing to buyouts and real estate [79], and it was reported to be selling about $2.5 billion of fund stakes [60]. George Suttles of the Commonfund Institute noted that endowments with more public equity have outperformed those heavy in alternatives in recent years [78].

Pensions are moving the other way. In March 2024 CalPERS raised its private equity target from 13% to 17% and private debt from 5% to 8%, taking private assets from 33% to 40% of the fund [80]. A defined-benefit pension has long liabilities and predictable outflows, so it is a reasonable holder of illiquidity. It also has a board that sees quarterly marks, which makes it the investor Asness expects to pay for smoothing [18].

The retail push

The newest buyers have the least capacity to wait. An August 2025 executive order directed the Labor Department to clarify how 401(k) plans may offer private assets, and the department proposed a rule in March 2026 [81]. Semi-liquid vehicles are the delivery mechanism. Morningstar counts about $607 billion in 567 evergreen funds as of March 2026 [82]. XA Investments counts 308 interval and tender-offer funds with about $233 billion of net assets at the end of 2025 [83]. Hamilton Lane expects evergreen funds to grow from about 5% of private markets to at least 20% within a decade [84].

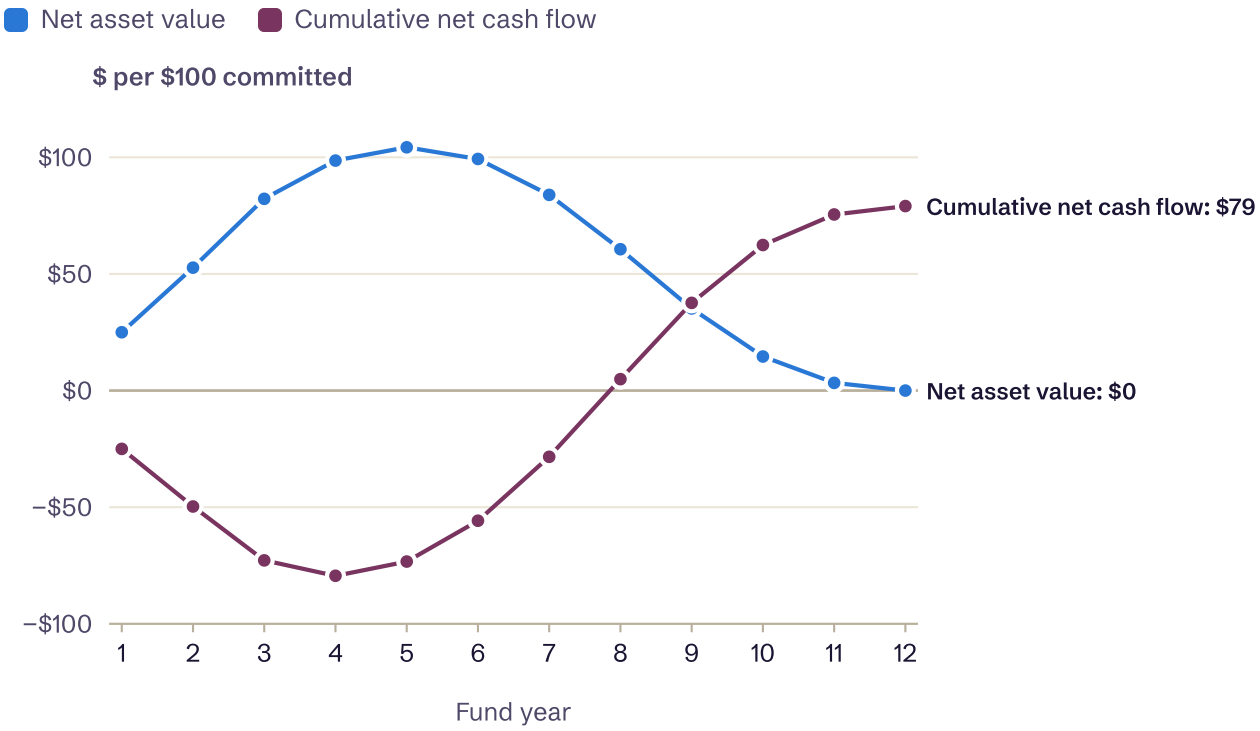
I am not against individuals owning private assets. I am against selling them a premium that requires patience in a wrapper that promises liquidity. The 2026 gates are what that mismatch looks like. An individual who needs cash in year three is the forced seller in Longstaff's model, and the investor most likely to redeem at the worst time is the one these products are built for.

Pacing and the J-curve

Owning illiquid assets is a cash-flow problem before it is a return problem. Figure 19 runs the Takahashi–Alexander model, which Yale built to forecast fund cash flows [85], for $100 committed to a 12-year fund. I assume capital is called at 25%, 33% and then 50% of the remaining commitment each year, assets grow 12% a year, and distributions follow the model's bow factor of 2.5. These are my assumptions, chosen to look like a typical buyout fund.

Figure 19. The J-curve: you fund it for four years before it funds you

Illustrative $100 commitment to a 12-year fund, Takahashi-Alexander model, no fees



My illustration. Contributions: 25%, 33% and then 50% of the unfunded commitment each year. Growth: 12% a year. Distribution bow: 2.5. Life: 12 years. Result: $179 distributed on $100 paid in, a 12% IRR.

Source: Model from Takahashi and Alexander (2002) [85]; parameters and calculation mine.

The investor is $79 out of pocket at the bottom of the J in year 4, breaks even in year 8, and ends with $179 back on $100 paid in, a 12% IRR. Two pacing rules follow from the same model. To hold private NAV steady at a target, annual commitments have to be about 15% of that target, because each vintage's NAV peaks near $104 per $100 committed and vintages overlap. And the investor carries unfunded commitments of about 1.75 times a year's commitments at all times.

For a $100 million portfolio holding 20% in private funds, that means committing about $3 million a year and keeping about $5 million ready for calls, before any stress. If distributions slow, as they have since 2022, the bottom of the J gets deeper and later, and the reserve needs to be larger. None of this is a forecast. The point is the shape: you fund the strategy for about four years before it funds you.

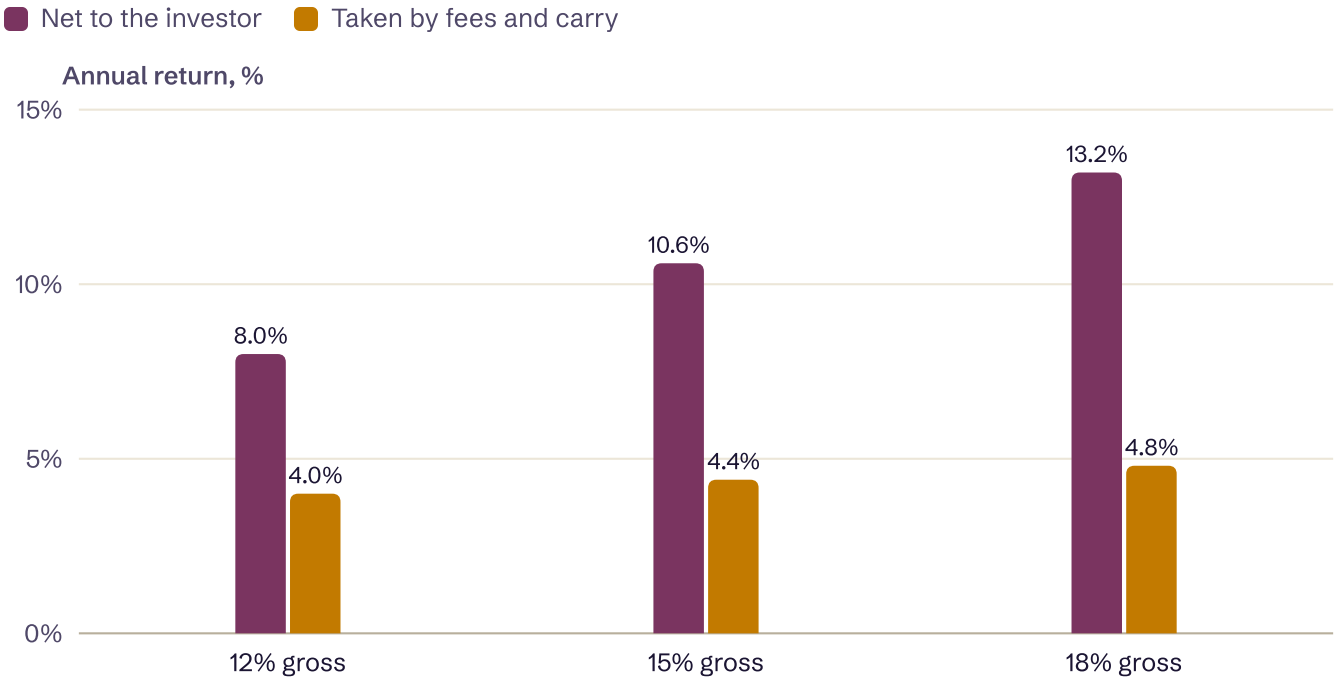
Fees

Fees are where the premium most often disappears. Metrick and Yasuda estimated that about two-thirds of a typical private equity manager's expected revenue comes from fixed fees that do not depend on performance [86]. Phalippou estimates that private equity funds have returned about 11% a year since 2006, roughly what public equity indexes returned, while paying about $230 billion of carried interest [27]. When CalPERS first disclosed carry in 2015, it reported $3.4 billion paid since 1990, against $24.2 billion of realized net gains [87].

My own stylized model shows the size of the drag. On the same J-curve cash flows, with a 2% fee on commitments for five years, 1.5% of NAV after that, and 20% carry over an 8% preferred return with full catch-up, a fund that earns 15% gross delivers 10.6% net, a gap of 4.4 points a year. At 12% gross the gap is 4.0 points and at 18% it is 4.8. Put another way, if public equities earn 10%, the fund has to earn about 14.4% gross just to tie them after fees. Fees consume most of a plausible premium before the investor sees any of it.

Figure 20. What fees take

Gross and net annual return in my stylized buyout fund, by gross return



My model, on the J-curve cash flows in Figure 19: a 2% fee on commitments for five years, 1.5% of NAV after that, and 20% carry over an 8% preferred return with full catch-up. AQR's independent estimate of all-in fees is 3.5 points a year [26].

Source: My calculation; fee terms as described. AQR [26].

Manager selection

Dispersion is the other side of the fee problem. In venture the gap between top- and bottom-quartile funds is close to 20 points of IRR [30]. Picking the top quartile is how most LPs justify their programs, but in buyouts persistence has weakened to the point that past top-quartile results say little about the next fund, while weak past results still predict weak future ones [23]. If you cannot reliably pick the top quartile, the quartile spread is risk, not opportunity.

8. The verdict

Assume zero

My position, after working in private markets and three versions of this paper: assume the illiquidity premium on a private fund is zero until someone shows otherwise, and insist on being shown it against a levered public benchmark, after fees. The burden of proof sits with whoever is selling the lockup.

Three questions decide it. Is there a premium, measured against a public portfolio with the same leverage and factor exposures? Is it larger than the fees? And am I the right holder, meaning I will not be a forced seller in the state where everyone else is selling too? For the average buyout, core real estate or infrastructure fund, the evidence in Section 4 answers no to the first question, and the other two stop mattering.

That leaves a short list. Secondaries bought from forced sellers, where the discount is the premium and the buyer is the one supplying liquidity. Senior direct lending when lenders have pricing power, which is after a year of redemptions and defaults and not before it [100]. Small buyouts with a manager whose record beats a levered small-cap index. Venture only with access to managers whose persistence you can document [23].

It rules out most of what is being sold in 2026: large-cap buyouts priced off public comparables, semi-liquid vehicles that charge private-market fees for a liquidity promise they cannot keep, continuation vehicles priced by the seller, and any strategy whose case rests on low reported volatility.

Be the liquidity

The practical version is to hold the option instead of selling it. Short inflation-protected Treasuries pay a positive real yield, 2.42% on the five-year on November 9, 2026 [98], so a reserve earns something while it waits. In a stress year that reserve buys fund stakes at 54% to 81% of NAV [51, 59], or lends on terms no one offered a year earlier [100]. The return on that reserve is the illiquidity premium. It is lumpy, it means doing nothing for years at a time, and it is the only version of the premium that the evidence in this paper supports.

How to underwrite it

If I were deciding on an illiquid commitment today, I would underwrite it against an explicit hurdle, built in four steps.

1. **Pick the right public benchmark.** For buyouts, a small-cap value index levered to the fund's debt. For direct lending, the leveraged loan index. For core real estate, REITs delevered to the fund's loan-to-value. The premium is measured against this, not against the S&P 500.
2. **Use true volatility.** Desmooth the track record or borrow a public proxy. Anson's numbers are a starting point: about 22% for buyouts, much more for early-stage venture, about 9% for direct lending [17].
3. **Price the lockup.** Longstaff's bound gives the most a perfectly timed seller could lose [4]. Scale it by your liquidity need, the share of that option you would realistically use. A perpetual endowment that never sells is near zero. An investor who may need the money is near one.
4. **Add the fees.** The gross hurdle a manager has to clear is the net premium plus the gross-to-net spread, which in my model runs 4 to 5 points for a standard buyout fund.

Table 3. Required net premium by lockup length

% a year over a public portfolio with matched risk, after fees. Liquidity need λ is the share of Longstaff's timing option the investor would use.

Lockup (years)	Upper bound, σ = 10%	Upper bound, σ = 22%	Upper bound, σ = 35%	Patient investor (λ = 25%, σ = 22%)	Stretched investor (λ = 75%, σ = 22%)
2	5.7%	12.8%	20.8%	3.4%	9.8%
3	4.6%	10.3%	16.5%	2.8%	7.9%
5	3.6%	7.8%	12.4%	2.2%	6.1%
7	3.0%	6.5%	10.2%	1.9%	5.1%
10	2.5%	5.4%	8.4%	1.6%	4.2%
12	2.3%	4.9%	7.5%	1.5%	3.9%

My calculation: required premium = $(1 + \lambda F)^{1/T} - 1$, where F is Longstaff's bound for lockup T and volatility σ. Upper-bound columns set λ = 100%. Add 4 to 5 points of fee drag for the gross hurdle. An illustration, not a forecast or advice.

Source: Longstaff (1995) [4]; volatility levels from Anson (2024) [17].

Table 3 runs the arithmetic. A patient investor, with a liquidity need of 25%, in a ten-year buyout fund needs about 1.6 points a year over a levered small-cap index, net of fees. A stretched investor at 75% needs 4.2 points. Add a 4.4-point fee drag and the gross hurdles become 6.0 and 8.6 points over the public equivalent. Few buyout managers have cleared the second number after adjusting for leverage and size, and the evidence in the earlier sections says the average manager has not cleared the first. The calculator below applies the same method to your own inputs.

What hurdle should an illiquid fund clear?

Move the inputs. The output is the extra return per year, over a public portfolio with the same risk, that pays you for the lockup.

Lockup	**10 years**	Volatility of the underlying asset	**22%**
Your liquidity need	**35%**	Fee drag, gross to net	**4.5% a year**

Most a perfectly timed seller could lose

40.7%

of value, Longstaff upper bound

Net premium you need

2.2%

a year over the public equivalent, after fees

Gross hurdle the manager must clear

6.7%

a year over the public equivalent, before fees

Method: $F = (2 + \sigma^2T/2)\cdot N(\sqrt{(\sigma^2T)/2}) + \sqrt{(\sigma^2T/2\pi)}\cdot e^{-\sigma^2T/8} - 1$ is Longstaff's bound on the value of being free to sell, as a share of value [4]; the first box shows it as a discount to the freely tradable price, $F / (1 + F)$. Liquidity need λ is the share of that option you would realistically use: near 0 for a perpetual endowment that never sells, near 100% for an investor who may be a forced seller. Net premium = $(1 + \lambda F)^{1/T} - 1$. Gross hurdle adds the fee drag. Use true (desmoothed) volatility, not reported volatility. An illustration, not advice.

Red flags in diligence

These are general patterns I learned to look for, not observations about any manager, fund or client.

- An IRR without a multiple, or a multiple without a PME. A strong IRR on a weak multiple usually means a credit line or early partial exits.
- No IRR shown without the subscription facility.
- Reported volatility below the volatility of the public companies the fund resembles. Ask for a lagged-beta estimate.
- Marks that barely fell in 2022, when public comparables fell 20% or more.
- DPI that comes from NAV loans, dividend recapitalizations or continuation vehicles rather than sales to outside buyers.
- A continuation vehicle priced at NAV, with the GP rolling its carry. Ask who set the price and whether a second bidder saw the asset.
- In semi-liquid funds: redemption terms that say “up to 5%” without saying what happens when requests exceed it, incentive fees on unrealized gains, and distributions funded by new subscriptions or borrowing.
- A benchmark chosen after the fact, especially the S&P 500 for a fund that owns small, levered companies.
- Attribution that cannot separate multiple expansion and leverage from operating improvement.

What changed my mind

Three things changed how I think about this since I wrote the first version.

The premium belongs to the holder, not the asset. The same buyout fund is a good investment for an endowment that will never sell and a bad one for a family that may need the money in year four. Diligence that ignores the investor's liquidity budget is incomplete.

Much of reported performance is measurement. In school I read about smoothing as a caveat. In practice it shapes allocations. It is why investors hold more private assets than an honest optimizer would give them, and why the 2022 marks took a year to arrive.

Private returns are less unique than they look. Stafford's replication [21], L'Her's benchmark [20] and the last ten years of the Cambridge data [24] say the same thing. Most of what private equity delivered can be bought in public markets with leverage and a value tilt. What cannot be bought publicly is real operating skill, which is rare, and the ability to supply liquidity when others need it, which is open to anyone patient enough to hold cash for it.

9. What I got wrong (and right) in 2025

The first version of this paper was a literature review for a finance course. Rereading it, some of it holds up and some does not.

What held up. The core claim, that illiquid assets must offer higher expected returns and that the premium has compressed where capital crowded in, is still right. So is the emphasis on Acharya and Pedersen's point that liquidity risk, not just illiquidity, is priced. The warning that naive optimizers over-allocate to smoothed assets, which I took from Dimitrov [90], looks better after 2022 than it did when I wrote it.

What I got wrong. I presented the traditional 3% to 5% private equity premium as an expectation. It is a hurdle, and the last decade did not clear it [24]. I cited a valuation rule of thumb that a two-year restriction implies a 25% to 35% discount [88]. Longstaff's upper bound for a typical buyout volatility is about 21%, and the secondary market priced buyout stakes at single-digit discounts in 2024 and 2025 [53, 54]. Discounts that large appear only in crises or in assets whose NAVs buyers do not trust.

I repeated State Street's finding that illiquidity improved pension outcomes by preventing bad timing [89] without the other side: the same lockup forces bad timing on investors who are overcommitted, as 2009 and 2022 showed. And I treated smoothed returns as a behavioral quirk, when they work more like a product feature that investors pay for [18].

What I missed. I gave the secondary market one clause, and it is now the best data we have on the price of illiquidity. I did not discuss fees at all, and they are the largest single reason the premium fails to reach investors. And I wrote about retail access to private markets as a policy debate. It is now a product category, with gates.

What I got wrong in the first rewrite. The earlier 2026 draft concluded that the premium was "real, but smaller than the fees." That was a hedge. A premium smaller than the fees is zero to the investor, and after leverage it is negative. I also treated leverage as a caveat to the Cambridge comparison. It accounts for the whole 25-year gap.

Sources

Public sources only. Where a figure comes from secondary coverage of a report I could not read directly, the entry says so, and the text marks approximate values. Figures labeled as my arithmetic are calculations on the cited numbers.

1. Amihud, Y., & Mendelson, H. (1986). Asset pricing and the bid-ask spread. Journal of Financial Economics, 17(2), 223-249. doi.org/10.1016/0304-405X(86)90065-6
2. Amihud, Y., & Mendelson, H. (1986). Liquidity and stock returns. Financial Analysts Journal, 42(3), 43–48. rpc.cfainstitute.org/research/financial-analysts-journal/1986/liquidity-and-stock-returns
3. Acharya, V. V., & Pedersen, L. H. (2005). Asset pricing with liquidity risk. Journal of Financial Economics, 77(2), 375–410. doi.org/10.1016/j.jfineco.2004.06.007
4. Longstaff, F. A. (1995). How much can marketability affect security values? Journal of Finance, 50(5), 1767-1774. doi.org/10.1111/j.1540-6261.1995.tb05197.x
5. Pástor, Ľ., & Stambaugh, R. F. (2003). Liquidity risk and expected stock returns. Journal of Political Economy, 111(3), 642–685. econpapers.repec.org/RePEc:ucp:jpolec:v:111:y:2003:i:3:p:642-685
6. Pástor, Ľ., & Stambaugh, R. F. (2019). Liquidity risk after 20 years. Critical Finance Review, 8, 277-299. NBER Working Paper 25774. www.nber.org/papers/w25774
7. Ben-Rephael, A., Kadan, O., & Wohl, A. (2015). The diminishing liquidity premium. Journal of Financial and Quantitative Analysis, 50(1-2), 197-229. ideas.repec.org/a/cup/jfinqa/v50y2015i1-2p197-229_00.html
8. Longstaff, F. A. (2004). The flight-to-liquidity premium in U.S. Treasury bond prices. Journal of Business, 77(3), 511–526. NBER Working Paper 9312. www.nber.org/papers/w9312
9. Krishnamurthy, A. (2002). The bond/old-bond spread. Journal of Financial Economics, 66(2–3), 463–506. static1.squarespace.com/static/5e6033a4ea02d801f37e15bb/t/5f624561a8d03f77b87e868f/1600275809874/krishnamurthy_bond_spread_jfe.pdf
10. Longstaff, F. A., Mithal, S., & Neis, E. (2005). Corporate yield spreads: Default risk or liquidity? New evidence from the credit default swap market. Journal of Finance, 60(5), 2213-2253. econpapers.repec.org/RePEc:bla:jfinan:v:60:y:2005:i:5:p:2213-2253
11. Chen, L., Lesmond, D. A., & Wei, J. (2007). Corporate yield spreads and bond liquidity. Journal of Finance, 62(1), 119-149. Summary: CFA Institute Digest (2007). rpc.cfainstitute.org/research/cfa-digest/2007/08/corporate-yield-spreads-and-bond-liquidity-digest-summary
12. Federal Reserve Bank of New York (2025, July). The Fed's Treasury purchase prices during the pandemic. Liberty Street Economics. libertystreeteconomics.newyorkfed.org/2025/07/the-feds-treasury-purchase-prices-during-the-pandemic
13. Kaplan, S. N., & Schoar, A. (2005). Private equity performance: Returns, persistence, and capital flows. Journal of Finance, 60(4), 1791–1823. web.mit.edu/aschoar/www/KaplanSchoar2005.pdf
14. Gredil, O., Griffiths, B., & Stucke, R. (2023). Benchmarking private equity: The direct alpha method. Journal of Corporate Finance, 81, 102360. doi.org/10.1016/j.jcorpfin.2023.102360
15. Schillinger, P., Braun, R., & Cornel, J. (2019). Distortion or cash flow management? Understanding credit facilities in private equity funds. Working paper, Technical University of Munich.
16. Getmansky, M., Lo, A. W., & Makarov, I. (2004). An econometric model of serial correlation and illiquidity in hedge fund returns. Journal of Financial Economics, 74(3), 529–609. NBER Working Paper 9571. www.nber.org/papers/w9571
17. Anson, M. (2024). Amortizing volatility across private capital investments. Journal of Portfolio Management, 50(7), 23–35. Figures as summarized by Alpha Architect, "The impact of amortizing volatility across private investments." www.pm-research.com/content/iijpormgmt/50/7/23
18. Asness, C. (2023, January 6). Volatility laundering. AQR Cliff's Perspectives. www.aqr.com/Insights/Perspectives/Volatility-Laundering
19. Phalippou, L., & Gottschalg, O. (2009). The performance of private equity funds. Review of Financial Studies, 22(4), 1747-1776. econpapers.repec.org/RePEc:oup:rfinst:v:22:y:2009:i:4:p:1747-1776
20. L'Her, J.-F., Stoyanova, R., Shaw, K., Scott, W., & Lai, C. (2016). A bottom-up approach to the risk-adjusted performance of the buyout fund market. Financial Analysts Journal, 72(4), 36–48. ideas.repec.org/a/taf/ufajxx/v72y2016i4p36-48.html
21. Stafford, E. (2022). Replicating private equity with value investing, homemade leverage, and hold-to-maturity accounting. Review of Financial Studies, 35(1), 299–342. papers.ssrn.com/sol3/papers.cfm?abstract_id=2720479
22. Harris, R. S., Jenkinson, T., & Kaplan, S. N. (2014). Private equity performance: What do we know? Journal of Finance, 69(5), 1851–1882. NBER Working Paper 17874. www.nber.org/papers/w17874
23. Harris, R. S., Jenkinson, T., Kaplan, S. N., & Stucke, R. (2023). Has persistence persisted in private equity? Evidence from buyout and venture capital funds. Journal of Corporate Finance, 81, 102361. NBER Working Paper 28109. www.nber.org/papers/w28109
24. Cambridge Associates (2026). US PE/VC benchmark commentary: Calendar year 2025. www.cambridgeassociates.com/insight/us-pe-vc-benchmark-commentary-calendar-year-2025
25. Ilmanen, A., Chandra, S., & McQuinn, N. (2020). Demystifying illiquid assets: Expected returns for private equity. Journal of Alternative Investments, 22(3), 8–30. www.aqr.com/insights/research/journal-article/demystifying-illiquid-assets-expected-returns-for-private-equity
26. AQR Capital Management (2026, January). 2026 capital market assumptions for major asset classes. Alternative Thinking, 2026 Issue 1 (Exhibits 6, A4 and A6). www.aqr.com/Insights/Research/Alternative-Thinking/2026-Capital-Market-Assumptions-for-Major-Asset-Classes
27. Phalippou, L. (2020). An inconvenient fact: Private equity returns and the billionaire factory. SSRN Working Paper 3623820. papers.ssrn.com/sol3/papers.cfm?abstract_id=3623820

28. Levine, S. (2014, August). Venture outcomes are even more skewed than you think (Correlation Ventures data). sethlevine.com/archives/2014/08/venture-outcomes-are-even-more-skewed-than-you-think.html
29. Evans, B. (2016). In praise of failure (Horsley Bridge data, 1985-2014). Reproduced at Climateer Investing. climateerinvest.blogspot.com/2016/08/power-law-return-characteristics-of.html
30. PitchBook (2023). 2023 Global fund performance report, as of Q4 2022. files.pitchbook.com/website/files/pdf/2023_Global_Fund_Performance_Report_as_of_Q4_2022_with_preliminary_Q1_2023_data.pdf
31. PitchBook (2026). Q4 2025 Global fund performance report. pitchbook.com/news/reports/q4-2025-global-fund-performance-report-with-preliminary-q1-2026-data
32. Cliffwater (2026, March 31). Cliffwater Direct Lending Index data supports strength of private credit (2025 results). www.cliffwater.com/files/docs/CliffwaterPressRelease_CDLI_2025_Results.pdf
33. Cliffwater (2023). 2023 Q2 report on U.S. direct lending. www.bdcs.com/docs/Cliffwater2023Q2ReportonUSDirectLending.pdf
34. Loomis Sayles (2026). Loomis on Loans: 2025 Morningstar LSTA U.S. Leveraged Loan Index return. www.loomissayles.com/?p=36470
35. Morgan Stanley Investment Management. The evolution of direct lending (citing PitchBook LCD). www.morganstanley.com/im/publication/insights/articles/article_evolutionofdirectlending.pdf
36. Reuters (2026, March 9). US private credit defaults hit record 9.2% in 2025, Fitch says. www.marketscreener.com/news/us-private-credit-defaults-hit-record-9-2-in-2025-fitch-says-ce7e5fd8df8fff2d
37. International Monetary Fund (2024, April). Global Financial Stability Report, Chapter 2: The rise and risks of private credit. www.imf.org/en/Publications/GFSR/Issues/2024/04/16/global-financial-stability-report-april-2024
38. Bank for International Settlements (2026). AI disruption in private credit: Exposure to software firms in BDCs. BIS Bulletin 128. www.bis.org/publ/bisbull128.htm
39. J.P. Morgan Asset Management (2026). Tech, software and BDCs: Navigating volatility and AI disruption in investment grade credit. am.jpmorgan.com/us/en/asset-management/institutional/insights/portfolio-insights/fixed-income/fixed-income-perspectives/tech-software-and-bdcs-navigating-volatility-and-ai-disruption-in-investment-grade-credit
40. Fortune (2025, October 15). Jamie Dimon issues private credit warning: “When you see one cockroach, there are probably more.” fortune.com/2025/10/15/jamie-dimon-issues-private-credit-warning-when-you-see-one-cockroach-there-are-probably-more
41. Institutional Real Estate, Inc. (2022). NCREIF: Strong performance continues for institutional real estate (NPI Q1 2022 5.33%, Q2 2022 3.23%). irei.com/news/ncreif-strong-performance-continues-institutional-real-estate-3
42. LaSalle Investment Management (2023, April). Q4 2022 U.S. private real estate returns review (NPI Q3 2022 0.6%, Q4 2022 –3.5%). www.lasalle.com/wp-content/uploads/2023/04/2022-q4-us-private-real-estate-returns-review.pdf
43. NCREIF (2024). NPI snapshot report, 4th quarter 2023. ncreif.org/__static/3d5760159a31b3731a6869303995366a/npi-snapshot-report-4q2023.pdf
44. NCREIF (2025). Expanded NPI snapshot, 4th quarter 2024. ncreif.org/__static/5c02e027491f1e8c4c29d6378aa5f68d/Expanded-NPI-Snapshot-4Q2024.pdf
45. RCLCO (2026, February). Q4 2025 NCREIF results summary. www.rclco.com/wp-content/uploads/2026/02/Q4-2025-NCREIF-Results-Summary.pdf
46. Capital Economics (2022). NCREIF Property Index (Q4 2021). www.capitaleconomics.com/clients/publications/us-commercial-property/us-commercial-property-data-response/ncreif-property-index-q4-2
47. Nareit (2022). REITs post second strongest year on record in 2021. reit.com/news/blog/market-commentary/reits-post-second-strongest-year-record-2021
48. Nareit. REIT industry fact sheets, December 2024 and December 2025 (FTSE Nareit All Equity REITs annual returns). www.reit.com/sites/default/files/2026-01/MediaFactSheet_Dec-2025.pdf
49. Andonov, A., Kräussl, R., & Rauh, J. (2021). Institutional investors and infrastructure investing. Review of Financial Studies, 34(8), 3880–3934. papers.ssrn.com/sol3/papers.cfm?abstract_id=3816820
50. Jefferies (2022, July). Global secondary market review, 1H 2022. www.jefferies.com/wp-content/uploads/files/IBBlast/1H2022-Jefferies-Global-Secondary-Market-Review.pdf
51. Jefferies (2023, January). Global secondary market review. www.jefferies.com/wp-content/uploads/files/IBBlast/Jefferies-Global_Secondary_Market_Review-January_2023.pdf
52. Jefferies (2024, January). Global secondary market review. www.jefferies.com/wp-content/uploads/sites/4/2024/01/Jefferies-Global-Secondary-Market-Review-January-2024.pdf
53. Jefferies (2025, January). Global secondary market review. www.jefferies.com/wp-content/uploads/sites/4/2025/02/Jefferies-Global-Secondary-Market-Review-January-2025.pdf
54. Jefferies (2026, January). 2025 global secondary market review: Another record-breaking year. www.jefferies.com/insights/the-big-picture/2025-global-secondary-market-review-another-record-breaking-year
55. Jefferies (2026, July). Global secondary market review, H1 2026. www.jefferies.com/wp-content/uploads/sites/4/2026/07/Jefferies-Global-Secondary-Market-Review-July-2026.pdf

56. Secondaries Investor (2024). GP-led deals surged through second half with secondaries set for big year: Jefferies. www.secondariesinvestor.com/gp-led-deals-surged-through-second-half-with-secondaries-set-for-big-year-jefferies
57. Lazard (2026). 2025 secondary market report; interim 2026 secondary market report. www.lazard.com/research-insights/lazard-2025-secondary-market-report
58. Evercore (2026, July). H1 2026 secondary market review. www.evercore.com/wp-content/uploads/2026/07/Evercore-H1-2026-Secondary-Market-Review.pdf
59. Nadauld, T. D., Sensoy, B. A., Vorkink, K., & Weisbach, M. S. (2019). The liquidity cost of private equity investments: Evidence from secondary market transactions. Journal of Financial Economics, 132(3), 158-181. NBER Working Paper 22404. www.nber.org/papers/w22404
60. Private Equity Wire (2025). Yale nears $2.5bn secondary sale of PE and VC stakes (citing Bloomberg). www.privateequitywire.co.uk/yale-nears-2-5bn-secondary-sale-of-pe-and-vc-stakes
61. Harvard Magazine (2009, September and November). Harvard endowment falls from $36.9 billion to $26 billion; $11 billion less. www.harvardmagazine.com/2009/09/sharp-endowment-decline-reported
62. Bain & Company (2025). Global private equity report 2025. www.bain.com/insights/outlook-is-a-recovery-starting-to-take-shape-global-private-equity-report-2025
63. Bain & Company (2026). Global private equity report 2026 and press release. www.bain.com/insights/outlook-gaining-traction-global-private-equity-report-2026
64. Bain & Company (2026). Private equity midyear report 2026. www.bain.com/insights/private-equity-midyear-report-2026
65. AltsWire (2022, December). Blackstone REIT limits redemptions. altswire.com/blackstone-reit-limits-redemptions
66. Blackstone Real Estate Income Trust (2024, March). Notice to stockholders; Commercial Observer (2024, March 1), BREIT fulfills 100 percent of redemption requests for first time since late 2022. www.breit.com/notice-to-stockholders-march-2024
67. InvestmentNews (2023, January). Blackstone's BREIT gets $4 billion boost from University of California. investmentnews.com/alternatives/blackstones-breit-gets-4-billion-boost-from-university-of-california/232318
68. Apollo Debt Solutions BDC (2026, March). Form 8-K, first-quarter repurchase offer. www.sec.gov/Archives/edgar/data/1837532/000119312526119949/d94402d8k.htm
69. Apollo Debt Solutions BDC (2026, September). Form 8-K, third-quarter repurchase offer; Reuters via DealStreetAsia, Apollo private credit fund redemption requests ease in third quarter. www.sec.gov/Archives/edgar/data/0001837532/000119312526398001/d142892d8k.htm
70. Ares Strategic Income Fund (2026). Form 424B3 prospectus supplement, share repurchase history. www.sec.gov/Archives/edgar/data/0001918712/000162828026054928/asifq2-202610qsupplement.htm
71. Blackstone Private Credit Fund (2026, June 4). Notice to shareholders; InvestmentNews, Blackstone BDC redemption surge signals a warning for other funds. www.bcred.com/bcred-notice-to-shareholders-6-4-2026
72. Bloomberg via Swissinfo (2026, June). Partners Group ready to cap more funds as withdrawals mount. www.swissinfo.ch/eng/partners-group-ready-to-cap-more-funds-as-withdrawals-mount/91525795
73. Alternative Credit Investor (2026, June 1). Fitch says non-traded BDCs can withstand redemptions surge. alternativecreditinvestor.com/2026/06/01/fitch-says-non-traded-bdcs-can-withstand-redemptions-surge
74. Cox Capital (2026, July 28). Press release on recurring liquidity program (Business Wire); Ares Strategic Income Fund (2026, October 2), Form 8-K on mini-tender offer. www.sec.gov/Archives/edgar/data/0001918712/000162828026064530/asif-20261002.htm
75. Preqin (2024). HSBC enters booming NAV finance market with debut fund (citing Oaktree's market estimate). www.preqin.com/news/hsbc-enters-booming-nav-finance-market-with-debut-fund
76. NACUBO (2026). U.S. higher education endowments report stable returns, increase spending to $33.4 billion in FY25. www.nacubo.org/Press-Releases/2026/US-Higher-Education-Endowments-Report-Stable-Returns-Increase-Spending-to-33-4-Billion-in-FY25
77. First Citizens Wealth (2026). Benchmark your university endowment (2025 NACUBO-Commonfund Study summary). www.firstcitizens.com/wealth/insights/business-planning/nacubo-endowment-study-benchmarks
78. ai-CIO (2026). In "turbulent year," university endowments report average 10.9% return. www.ai-cio.com/news/turbulent-year-university-endowments-report-average-return
79. Yale University (2025, October 24). Yale reports investment return for fiscal 2025; ai-CIO coverage. news.yale.edu/2025/10/24/yale-reports-investment-return-fiscal-2025
80. CalPERS (2024, March). CalPERS will increase private markets investments. www.calpers.ca.gov/newsroom/calpers-news/2024/calpers-will-increase-private-markets-investments
81. U.S. Department of Labor (2026, March 30). Proposed rule on fiduciary duties in selecting designated investment alternatives, following Executive Order 14330 (August 7, 2025). www.dol.gov/newsroom/releases/ebsa/ebsa20260330
82. Dakota (2026). Top 10 evergreen funds in private markets, Q2 2026 (citing Morningstar). www.dakota.com/resources/blog/top-10-evergreen-funds-in-private-markets-q2-2026
83. XA Investments (2026). Fourth quarter 2025 market update on non-listed closed-end funds. finviz.com/news/277862/xa-investments-reports-record-growth-in-2025-in-its-fourth-quarter-2025-market-update

84. InvestmentNews (2025). Private markets face some “recalibration” but evergreen funds set to lead growth (Hamilton Lane 2025 Market Overview). www.investmentnews.com/industry-news/private-markets-face-some-recalibration-but-evergreen-funds-set-to-lead-growth/259666
85. Takahashi, D., & Alexander, S. (2002). Illiquid alternative asset fund modeling. Journal of Portfolio Management, 28(2), 90–100.
86. Metrick, A., & Yasuda, A. (2010). The economics of private equity funds. Review of Financial Studies, 23(6), 2303-2341. rodneywhitecenter.wharton.upenn.edu/wp-content/uploads/2014/04/0717.pdf
87. Fortune (2015, November 24). CalPERS finally details its private equity manager profits. fortune.com/2015/11/24/calpers-finally-details-its-private-equity-manager-profits
88. Barick, G. (2024). Pricing the illiquidity premium for private market assets. International Journal for Multidisciplinary Research, 6(1), 1–12. (Cited in the 2025 version.) ijfmr.com
89. Fink, A. M. (2019). In praise of illiquidity. Fiduciary Insights, State Street Global Advisors. (Cited in the 2025 version.) www.ssga.com
90. Dimitrov, D. (2025). Untangling illiquidity: Optimal asset allocation with private asset classes. DNB Working Paper No. 827, De Nederlandsche Bank. (Cited in the 2025 version.) www.dnb.nl
91. Franzoni, F., Nowak, E., & Phalippou, L. (2012). Private equity performance and liquidity risk. Journal of Finance, 67(6), 2341-2373. Figures from the working paper version. www.netspar.nl/wp-content/uploads/024_Nowak_revised.pdf
92. Boyer, B., Nadauld, T. D., Vorkink, K. P., & Weisbach, M. S. (2021). Discount rate risk in private equity: Evidence from secondary market transactions. NBER Working Paper 28691, later published in the Journal of Finance. Figures from the working paper. www.nber.org/papers/w28691
93. French, K. R. (2026). Data library: Fama/French 3 factors, monthly, CRSP data through September 2026. Used for the U.S. stock market return and the one-month Treasury bill return. mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
94. Bary, A. (2026, February 25). Private-credit BDC market offers a choice: Cheap public funds or richer private ones. Barron’s (citing Raymond James), via Tiger Brokers news. www.itiger.com/news/2614723033
95. Scottish Financial News (2026, August 28). CT Private Equity Trust share price falls 9.3% as discount to NAV hits 28.9%. www.scottishfinancialnews.com/articles/ct-private-equity-trust-share-price-falls-93-as-discount-to-nav-hits-289
96. interactive investor (2026, September 4). Discount Delver: the 10 cheapest trusts on 4 September 2026. www.ii.co.uk/analysis-commentary/discount-delver-10-cheapest-trusts-4-september-2026-ii540184
97. Reuters via The Standard (2026, July 2). Blue Owl keeps withdrawal limits even as redemption pressure modestly eases. www.thestandard.com.hk/finance/article/336246/Blue-Owl-keeps-withdrawal-limits-even-as-redemption-pressure-modestly-eases
98. U.S. Department of the Treasury (2026). Daily Treasury par real yield curve rates, December 31, 2025 and November 9, 2026. home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_real_yield_curve&field_tdr_date_value=2026
99. Ang, A., Papanikolaou, D., & Westerfield, M. M. (2014). Portfolio choice with illiquid assets. Management Science, 60(11), 2737-2761. NBER Working Paper 19436 (Table 3). www.nber.org/papers/w19436
100. PIMCO (2026). The credit market lens: Green shoots and legacy risks in direct lending (citing PitchBook, Lincoln International and KBRA DLD; data as of September 10, 2026). www.pimco.com/us/en/insights/the-credit-market-lens-green-shoots-and-legacy-risks-in-direct-lending
101. Octus (2026). BDCs trading at 50% discount to NAV, but nonaccruals ticking downward. octus.com/resources/blog/bdcs-trading-at-50-discount-to-nav-but-nonaccruals-ticking-downward